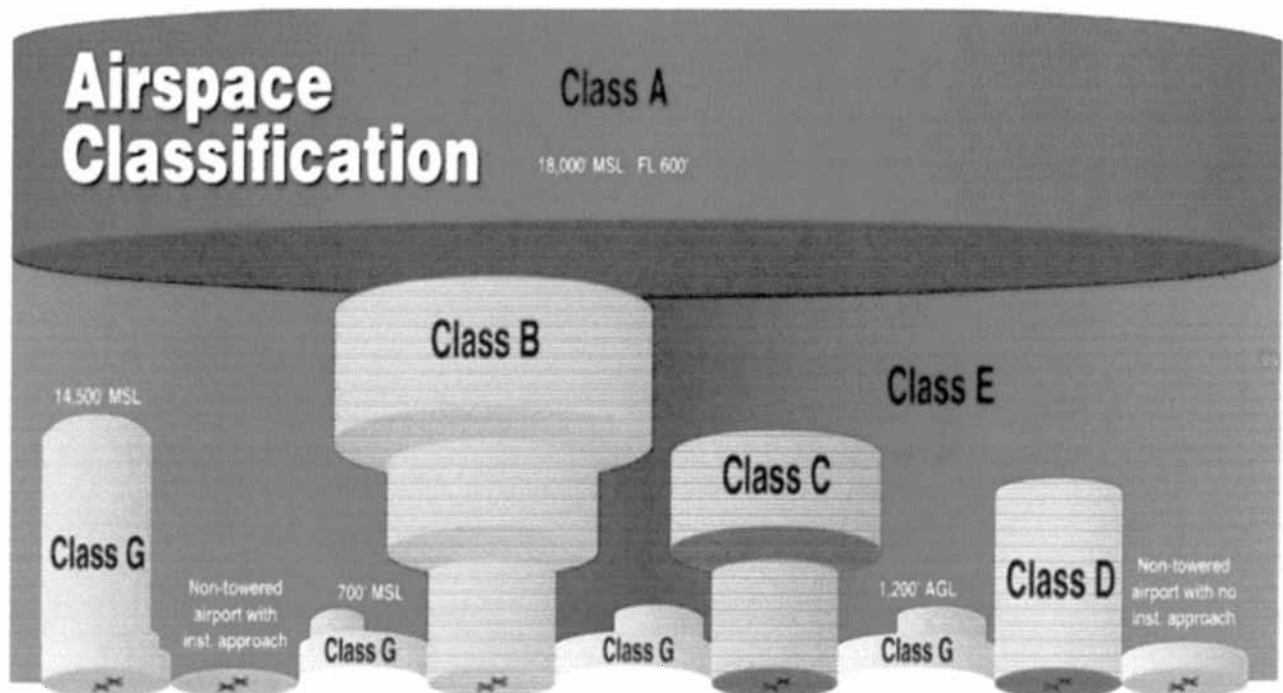
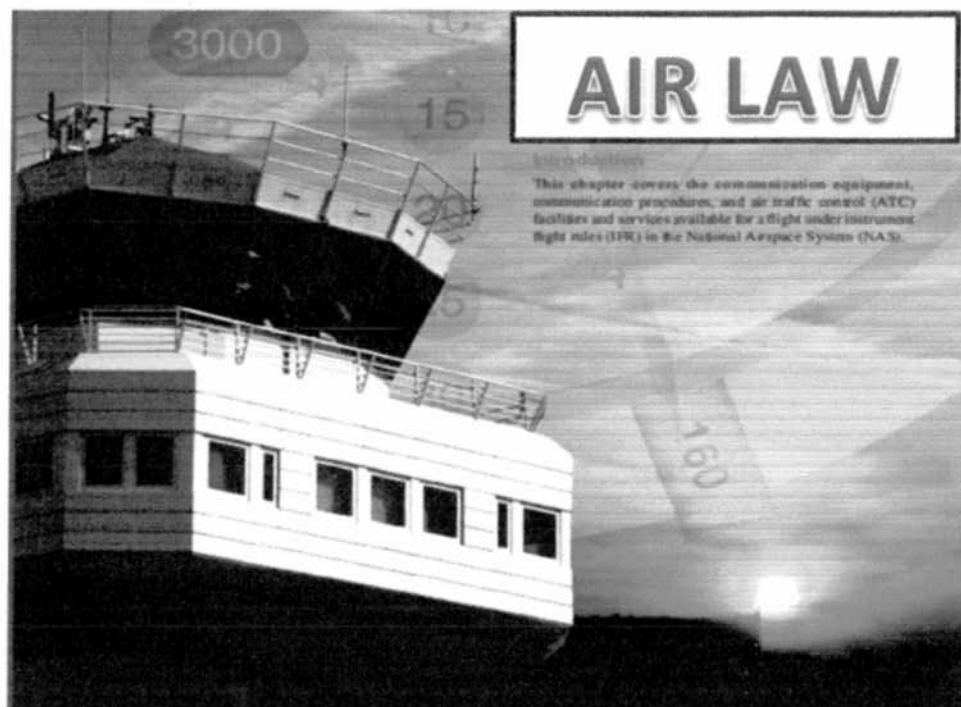




CAPT. SAHIL KHURANA
SEC-7, DWARKA, NEW DELHI-75
(+91) 9871866290, 9971663490



AIR LAW

RULES OF AIR – ANNEX 2

Annex 1	PERSONNEL LICENSING
Annex 2	RULES OF THE AIR
Annex 3	METEOROLOGICAL SERVICE FOR INTL AIR NAVIGATION
Annex 4	AERONAUTICAL CHARTS
Annex 5	UNITS OF MEASUREMENT AIR GROUND OPS
Annex 6	OPERATION OF AIRCRAFT
Annex 7	A/C NATIONALITY AND REGISTRATION MARKS
Annex 8	AIRWORTHINESS OF AIRCRAFT
Annex 9	FACILITATION
Annex 10	AERONAUTICAL TELECOM
Annex 11	AIR TRAFFIC SERVICES
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Annex 14	AERODROMES
Annex 15	AERONAUTICAL INFORMATION SERVICES
Annex 16	ENVIRONMENTAL PROTECTION
Annex 17	SECURITY SAFEGUARDING INTL. CIVIL AVIATION FROM UNLAWFUL INTERFERENCE
Annex 18	SAFE TRANSPORT DANGEROUS GOODS BY AIR

AEROBATIC FLIGHT - *Manoeuvres intentionally performed by an aircraft involving an abrupt change in its attitude, an abnormal attitude, or an abnormal variation in speed.*

MOVEMENT AREA - *That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).*

MANOEUVRING AREA - *That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.*



AEROBATIC FLIGHT -NO PERSON SHALL FLY AEROBATICALLY

- ☐ *In the vicinity of an aerodrome at a distance of less than 2 NM from the nearest point of the perimeter of the aerodrome unless being flown higher than 1800m or 6000 feet.*
- ☐ *When flying over city town village or populated area except when written permission by DG.*
- ☐ *Aerobatics should commence at such a height so it is to be completed at a height not less than 2000 feet above AGL.*
- ☐ *By a licensed pilot or accompanied by a licensed pilot Instructor.*
- ☐ *Over aerodrome Minimum Height – 6000 feet Agl.*

PROHIBITED AREAS AND RESTRICTED AREAS

Aircraft shall not be flown in a prohibited area, or in a restricted area, the particulars of which have been duly published, except in accordance with the conditions of the restrictions or by permission of the State over whose territory the areas are established.

- ☐ *If a following signal given by ground indicate –*
- ☐ *“ A series of projectiles discharged at intervals of 10 seconds, showing on bursting Red and Green lights.”*
- ☐ *Indicates - The Aircraft is flying in the vicinity of a RESTRICTED, PROHIBITED or DANGER area and should take necessary actions.*
- ☐ *Necessary actions includes –*
- ☐ *FIRSTLY – Will not fly further into the Prohibited Area.*
- ☐ *SECONDLY – Will not descend while still above prohibited Area.*
- ☐ *THIRDLY- Will ASA he is aware of the fact , give the signal of distress MAYDAY ,MAYDAY, MAYDAY by R/T*
- ☐ *FOURTHLY- Immediately Squawk Distress code “7700” and land to the nearest aerodrome outside Prohibited Airspace.*



Restricted area- An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is restricted in accordance with certain specified conditions.

Prohibited area- An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is totally prohibited.

PROHIBITED AREAS IN INDIA

BARODA	---- UNLIMITED
BOMBAY (Tower of Silence)	1mile UNLIMITED
MATHURA REFINERY	10kms UNLIMITED
RASTHTRAPATI BHAWAN	---- UNLIMITED
BHUBANESHWAR TOWER	---- 50000'

Danger area- An airspace of defined dimensions within which activities dangerous to the flight of aircraft may exist at specified times.

FORMATION FLIGHTS

Aircraft shall not be flown in formation except by prearrangement among the pilots-in-command of the aircraft taking part in the flight and, for formation flight in controlled airspace, in accordance with the conditions prescribed by the appropriate ATS authority

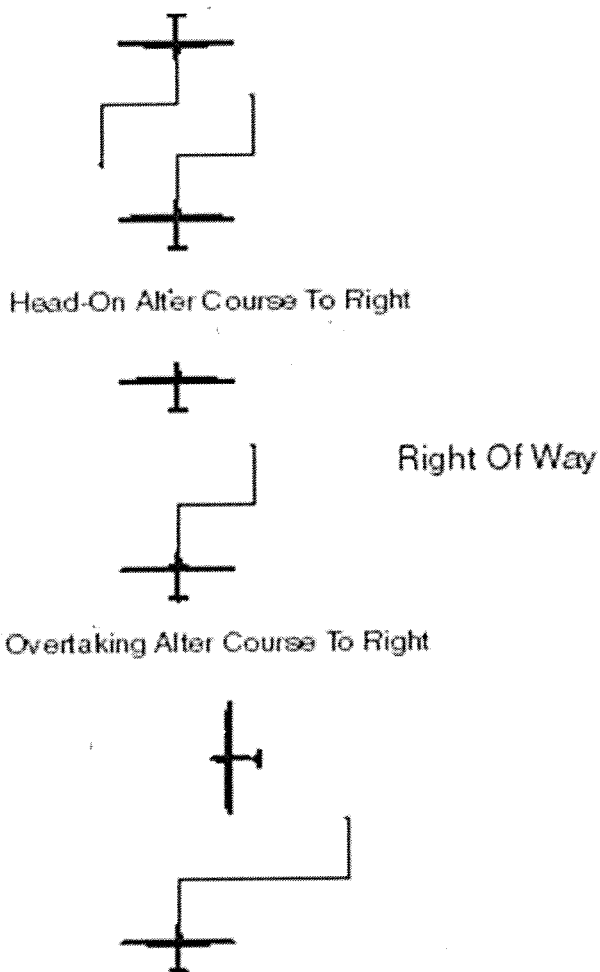
No formation Flights for Civil Aircraft

AVOIDANCE OF COLLISION RULES

Right of Way means the right to proceed without alteration of course. The aircraft that has the right of way is required to maintain its heading and speed, and observe the other aircraft whilst the collision risk exists.

Right of way Approaching head-on - When two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

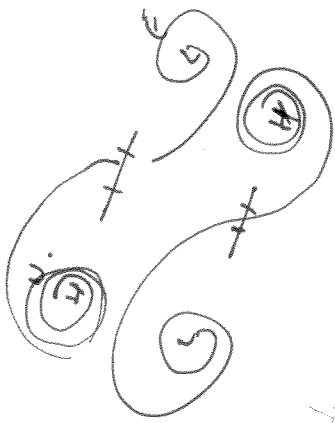
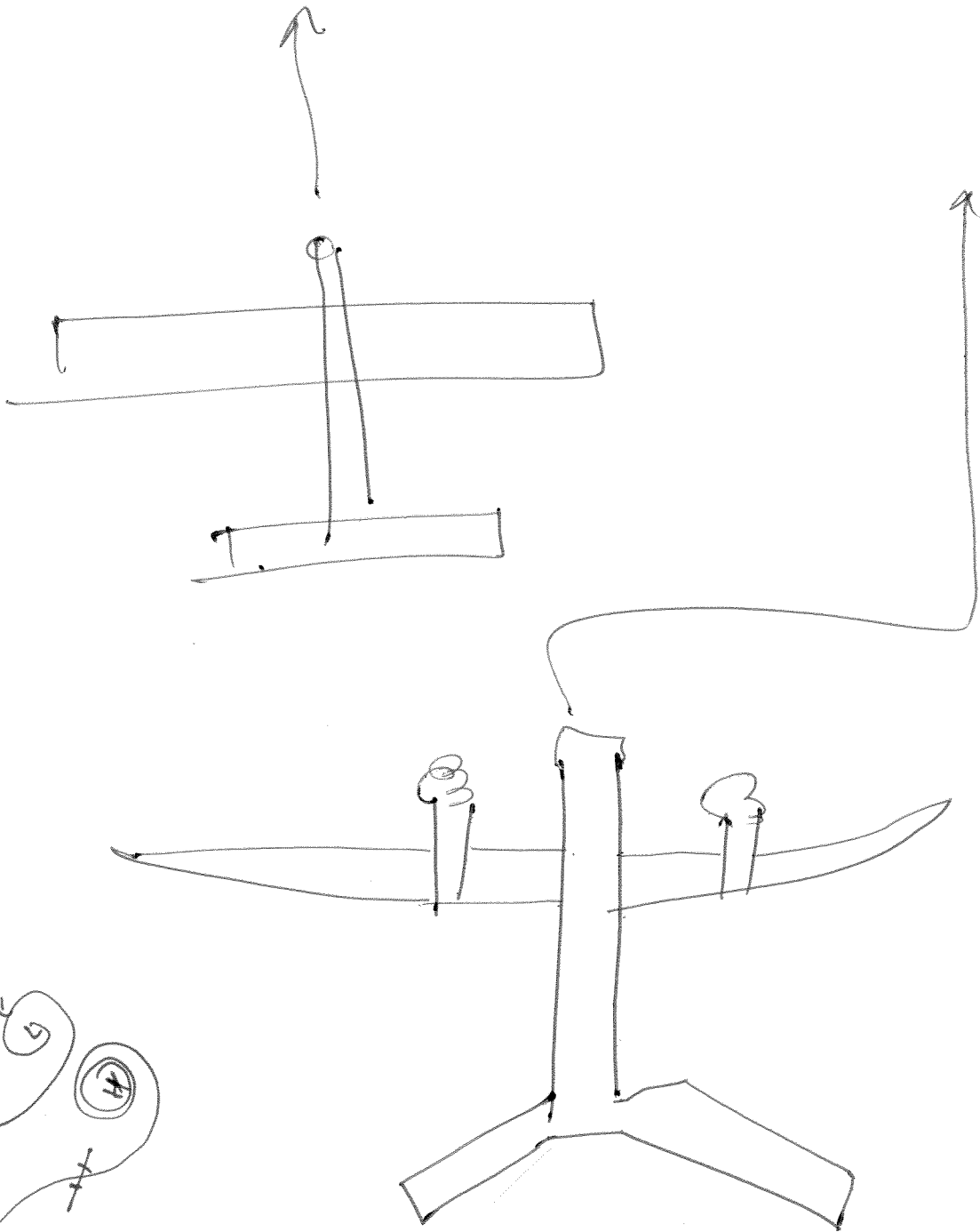
Converging - When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way, except as follows



GENEREAL RULES FOR AIR COLLISION AVOIDANCE

Power driven flying machines must give way to airships, balloons, gliders. Airships must give way to balloons and gliders. Gliders must give way to balloons and power driven flying machines must give way to towing aircrafts.

IMP : The A/C which has other aircraft on its right shall give way and the aircraft on the right has a right of way



Points →

- ① A/c on the Runway will have priority over a/c on taxiway.
- ② A/c on final Approach will have priority over a/c in flight.

- ③ A/c with Emergency on Board will have priority over any other flight.

Long finds → SNM

L 177
L 176

OVERTAKING

An aircraft that is being overtaken has the right of way and the overtaking aircraft, whether climbing, descending, or in horizontal flight, shall keep out of the way by altering its heading to the right.

An overtaking aircraft is an aircraft that approaches from the rear on a line forming an angle of less than 70°.

LANDING

An aircraft in flight, or operating on the ground or water, shall give way to aircraft landing or in the final stages of an approach to land.

SURFACE MOVEMENT OF AIRCRAFT

In case of danger of collision between two aircraft taxiing on the movement area of an aerodrome the following shall apply:

- ☐ *When two aircraft are APPROACHING HEAD ON, or approximately so, each shall stop or where practicable alter its course to the right so as to keep well clear.*
- ☐ *When two aircraft are on a CONVERGING COURSE, the one which has the other on its right shall give way.*
- ☐ *An aircraft which is being OVERTAKEN by another aircraft shall have the right-of-way and the overtaking aircraft shall keep well clear of the other aircraft.*

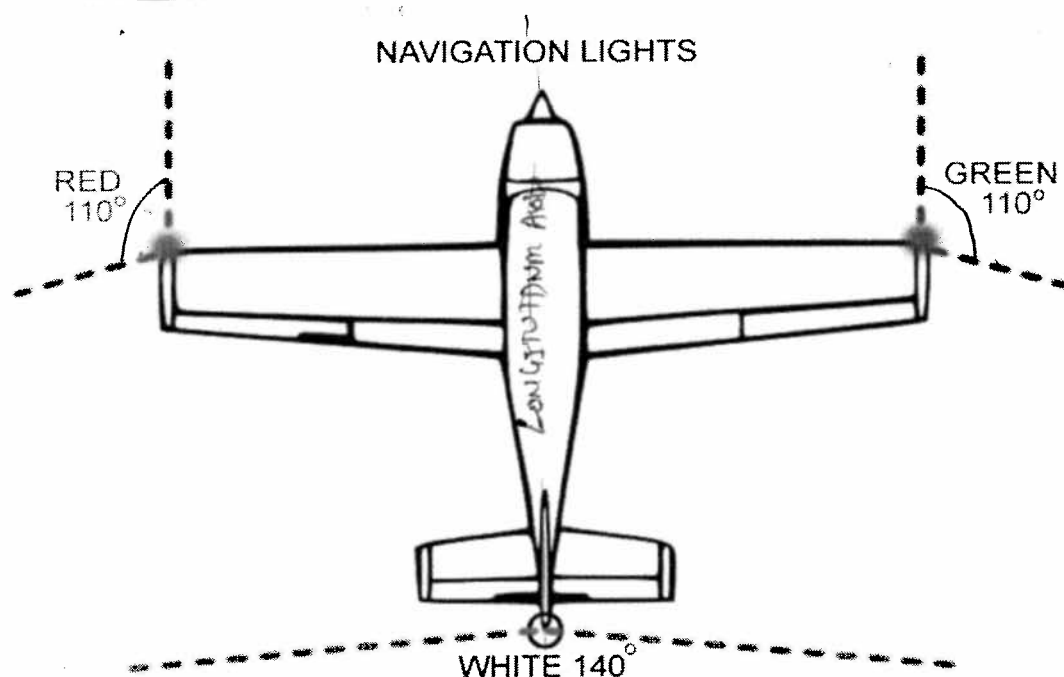
LIGHTS TO BE DISPLAYED BY AIRCRAFT

India's Rule 30 min after Sunset. 30 min before Sunrise

When Lights must be Displayed From sunset to sunrise (or during any other period prescribed by the appropriate authority), all aircraft in flight or on the movement area of an aerodrome must display:

- ☐ Anti collision lights intended to attract the attention of other aircraft.
- ☐ Navigation lights intended to indicate the relative path of the aircraft to an observer. No other lights shall be displayed if they are likely to be mistaken for the navigation lights.

- Unless stationary, and otherwise adequately illuminated, all aircraft on the movement area of an aerodrome shall display lights intended to indicate the extremities of their structure and to attract attention to the aircraft. (1)
- All aircraft operating on the movement area of an aerodrome whose engines are running shall display lights which indicate that fact.



The angle of coverage : Parallel to the longitudinal axis to 110° left and other 110° to the right.

The angle of coverage : Parallel to the longitudinal axis to 70° either side.

ANTI-COLLISION LIGHT

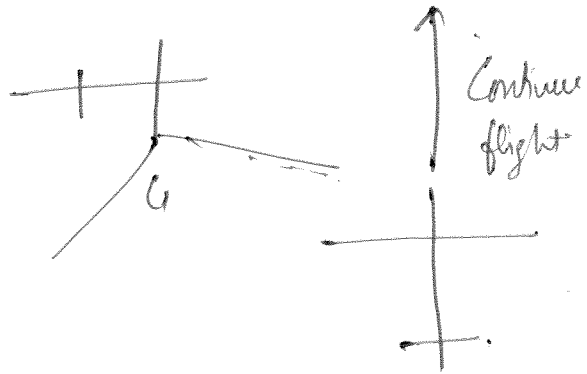
Anti-Collision Lights

All aircraft, in flight or operating on the movement area of an aerodrome, that are fitted with anti collision

lights, shall display these lights at all times. If an anti-collision light fails in flight, the light is to be repaired prior to the next flight.

Ques On a Relative Bearing of 270° . You see a Green light of another A/c. You will

- (A) Continue the flight.
- (B) Turn right
- (C) Turn left



OLS PILOT'S SAYING

Green to Green, all serene.

Red to Red, go ahead.

Red to Green, you must be seen

Green to Red, you could end up dead

Failure of Lights

- ☐ When a pilot is aware that a navigation light has failed, ATC is to be informed and the aircraft is to land and have the light repaired before continuing the flight.
- ☐ When Anti Collision light fails Aircraft can continue flight and repair it before next flight...Since it is not by law.

FLIGHT PLAN RULE

IS NEEDED FOR

- ☐ Any flight or portion of a flight requiring an ATC service.
- ☐ Any IFR flight in advisory airspace.
- ☐ Any flight where the appropriate ATC service is required to provide a flight information service (FIS), alerting service, and search and rescue (SAR) service;
- ☐ Where the authority has determined that a FPL should be filed to facilitate coordination between civilian and military authorities;
- ☐ Any flight across international borders.

60 minutes

For a flight to be provided with an ATC service or advisory ATC, the FPL is to be filed at least 60 minutes before departure.

10 minutes

For the filing of a FPL in flight, the FPL is to be filed at least 10 minutes before the aircraft is estimated to reach:

- ☐ The intended point of entry into an area where A TC advisory is to be provided; or
- ☐ The point of crossing an airway or advisory route.

DELAYS AFTER FILING

In the event of a delay of 30 minutes in excess of the estimated off-block time for a controlled flight or a delay of one hour for an uncontrolled flight for which a flight plan has been submitted, the flight plan should be amended or a new flight plan submitted and the old flight plan cancelled.

Adherence to the FPL

A pilot operating a controlled flight is required to adhere to the CPL (Current Flight Plan)

Inadvertent Changes

If a controlled flight deviates from the FPL, the following action is to be taken:

- ① ☐ Deviation from Track

Adjust the heading to regain the desired track as soon as practicable.

- ② ☐ Variation in TAS

If TAS at cruising level changes (or is expected to change) by 5% or more from that given in the FPL, inform the ATC.

- ③ ☐ Change in ETA

If the ETA changes by more than +/-3 minutes, the ETA is to be revised to ATC. In an Oceanic Control Area, report changes in ETA of +/- 3 mins or more.

LIGHT SIGNAL

Steady RED

A/c in flight.

Do not Land
Continue Circling.

A/c on ground

stop

FLASHING RED

Do not Land
Aerodrome unsafe/closed.

move clear of
landing area.

FLASHING GREEN

Return to Aerodrome
and await Landing
clearance.

clear to taxi

Steady Green

Return to Aerodrome
and clear to land

Clear to take-off.

Flashing white
Green

Return to Aerodrome
Land on Receipt of steady
Green

Return to starting point
of Aerodrome

Irregular switching on/off of Nav lights and Landing lights. Indicate that the A/c has Emergency and urgency to Land.

To Acknowledge the signals A/c may Rock his wings.

RED Pyrotechnique (Red flame) → Do not withstand with prior instruction

* Ifr flight with unlimited visibility (VMC) after Comm. failure will Land to nearest Aerodrome maintaining VMC and inform ATC.

Time

In all communications, time is to be expressed as Co-ordinated Universal Time (UTC) utilising the 24 hour clock. A time check is to be obtained before operating a controlled flight.

COMMUNICATIONS RULE

- ☐ *A controlled flight is required to maintain two way RTF communications with the controlling ATCU.*
- ☐ *Communication Failure in VMC*

If an aircraft is unable to communicate (receive and acknowledge ATC instructions and indicate a state of emergency), in addition to squawking 7600 and maintaining a visual watch for signals, if flying in VMC, maintain VMC and land at the nearest suitable aerodrome. ATC is to be informed as soon as possible once the aircraft lands.

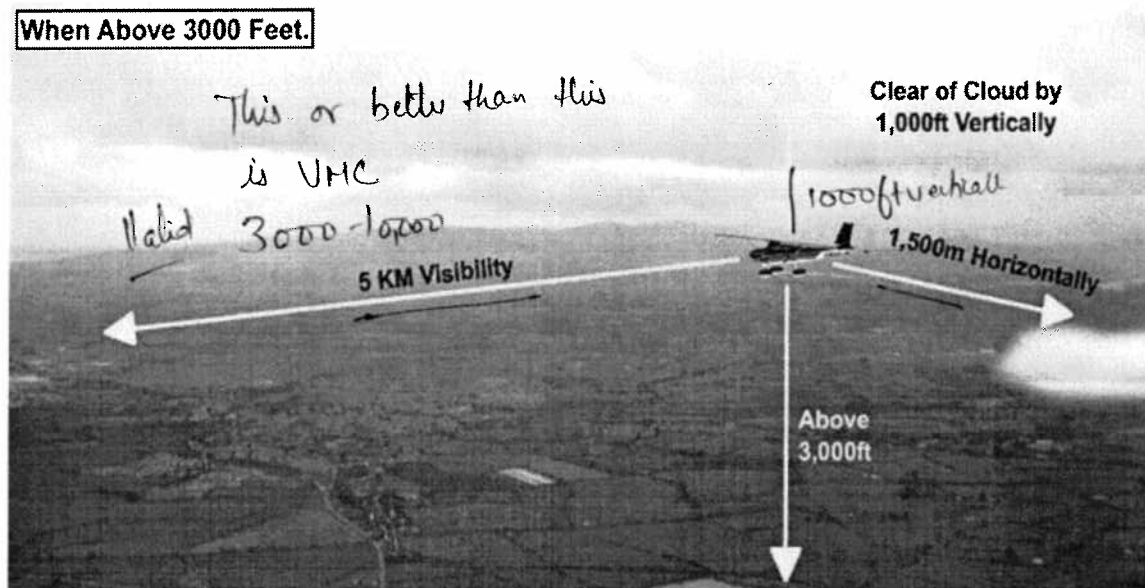
Communication Failure in IMC

- ☐ *Maintain the last assigned speed and/or level, for a period of 20 minutes after the failure to report over the last compulsory reporting point;*
- ☐ *Proceed in accordance with the FPL to the navigation facility serving the destination aerodrome and hold on that facility;*
- ☐ *Descend from the facility at the last received and acknowledged Expected Approach Time (EAT), or where no EAT has been issued, at the ETA from the FPL (4); then*
- ☐ *Fly a normal instrument approach;*
- ☐ *Land within 30 minutes of the ETA.*

Estimated time of arrival(ETA) - For IFR flights, the time at which it is estimated that the aircraft will arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced.

Expected approach time (EAT) - The time at which ATC expects that an arriving aircraft, following a delay, will leave the holding point to complete its approach for a landing.

RULES OF THE AIR FLIGHT RULES –VFR AND IFR



IMC [Instrument meteorological conditions] - Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, less than the minima specified for visual meteorological conditions.

Special VFR flight - A VFR flight cleared by air traffic control

to operate within a control zone in meteorological conditions below VMC.

VMC [Visual meteorological conditions] - Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, equal to or better than specified minima.

Visibility - Visibility for aeronautical purposes is the greater of:

- The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;
- The greatest distance at which lights in the vicinity of 1 000 candles can be seen and identified against an unlit background.

Ceiling - The height above the ground or water of the base of the lowest layer of cloud below 6 000 metres (20 000 feet) covering more than half the sky.

Class of Airspace	Vertical Limits	Visibility	Distance from Cloud
A, B, C D and E and F and G (Above 3 000ft AMSL or above 1 000ft above terrain, whichever is higher)	At and above FL 100 ⁽¹⁾	8 km	1000ft vertical, and 1500m horizontal from cloud
	Below FL 100	5 km	
F and G At and below 3 000ft AMSL or 1 000ft above terrain whichever is the higher		5 km ⁽²⁾	Clear of cloud and in sight of the surface

1000ft vertical
and 1500m
horizontal
from cloud

TAKE-OFF AND LANDING

VFR flights cannot take-off or land at an aerodrome in a CTR, or enter an aerodrome traffic zone or traffic pattern when:

- ❖ The ceiling is less than 1500 ft
- ❖ The visibility is less than 5 km

NIGHT – NO VFR AT NIGHT

MINIMUM HEIGHTS

A VFR flight shall not be flown:

- ☐ Over the congested areas of cities, towns or settlements, or over an open air assembly of persons, at a height less than 300 m (1000 ft) above the highest-obstacle within a radius of 600 m from the aircraft.
- ☐ Elsewhere, at a height less than 150 m (500 ft) above the ground or water.

Weather Deterioration below VMC

If it becomes evident that a controlled VFR flight will not remain in VMC, the pilot is to:

- ☐ Request an amended clearance to continue to the destination aerodrome by another route remaining in VMC;
- ☐ Land at the nearest useable aerodrome; or
- ☐ If operating in a CTR, request a SVFR clearance; or
- ☐ File an IFR FPL.

SPECIAL VFR (SVFR)

SVFR is defined as a clearance to fly within a CTR in conditions less than VMC in which the pilot remains clear of cloud and in visual contact with the ground.

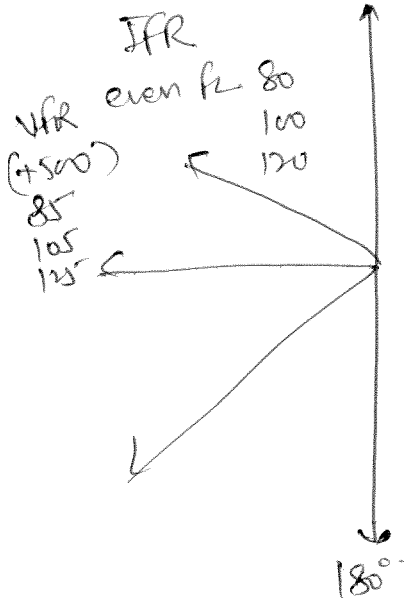
Provision of SVFR

Where a pilot cannot, or has good reason not to, comply with IFR in a CTR he/she may request a special VFR (SVFR) clearance to:

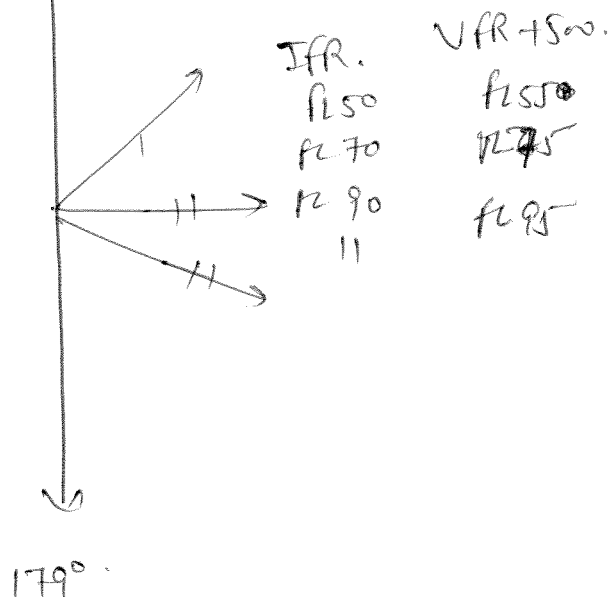
- Enter a CTR to land at an aerodrome within the CTR;
- Take off from an aerodrome within a CTR and depart from the CTR; or
- To fly between aerodromes within a CTR.

Take-off Conditions

359 (180-359°)



000 (magnetic Tracks)



According to ICAO a SVFR flight may take off from an aerodrome in a CTR providing the ground visibility is not less than 1500 m

MINIMUM HEIGHT RULE (IMC)

- Over high terrain or in mountainous areas, at a level which is at least 600 m (2 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft;
- Elsewhere than as specified in a), at a level which is at least 300 m (1 000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft

IFR is Flown when ever weather is below VMC and when Pilot wishes to fly IFR but with proper navigational equipment's on board

Reduced Vertical Separation Minima (RVSM)

The desired cruising levels for turbo-jet aircraft are adjacent to the tropopause (typically FL350 -370). In the upper airspace these levels quickly become occupied and congestion arises.

To overcome this (in part), a system of reduced vertical separation is applied whereby the 1000 ft separation between FLs is maintained up to FL410. Above FL410, the inaccuracy in the altimeter operation is too great for continuation of the reduced minima. This immediately doubles the available FLs between FL290 and the limit of the application, FL410.

This standard is known as Reduced Vertical Separation Minima (RVSM)

REQUIREMENTS

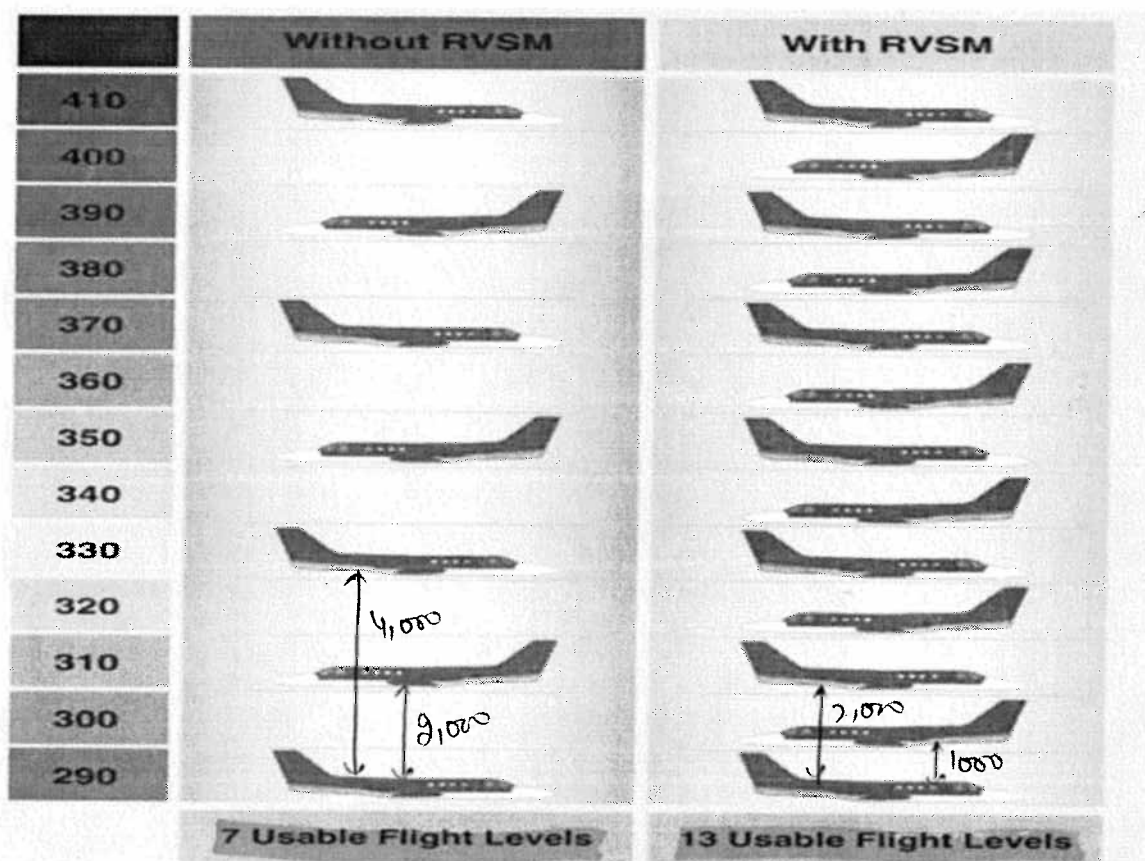
To fly in airspace where RVSM is applied, an aircraft must be equipped with –

- ☐ Two independent altitude measuring systems;
- ☐ An altitude alerting system (activated by deviation from the selected altitude);
- ☐ An automatic altitude control system (height lock),

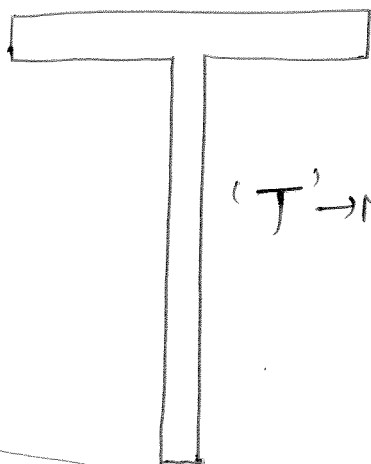
- 3] RVSM Airspace starts from FL290 to FL410
- 3] minimum separation Above FL290 is 2,000ft
- 3] minimum separation Between Same Track Alt Above FL290 is 1000ft
- 4]. minimum separation Above FL290 in RVSM Airspace is 1000ft.
- 5]. minimum separation in RVSM Above FL290 for Same Track is 2,000ft.
- 6]. minimum separation Below FL290 is 1000ft opposite traffic and 2,000 ft (Same Side)

- ☐ An SSR system with altitude reporting (mode C) connected to the system used for the automatic altitude control system.
- ☐ Additionally, the operator must be approved for RVSM operations.

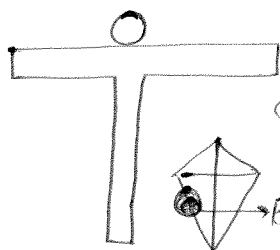
Reduced Vertical Separation Minima (RVSM)



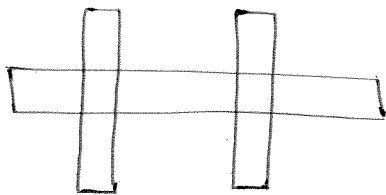
Reduced vertical separation minima or minimum (RVSM) is the reduction of the standard vertical separation required between aircraft flying between FL290 (29,000 ft) and FL410 (41,000 ft) inclusive, from 2,000 feet to 1,000 feet. This therefore increases the number of aircraft that can safely fly in a particular volume of airspace.



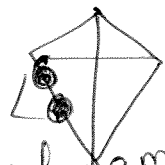
'T' → Means Landing in the direction of 'T'.



'T' with Disc → Landing and Take off may not coincide in same direction.
Black ball hanging from mast

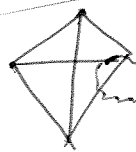


double white cross →

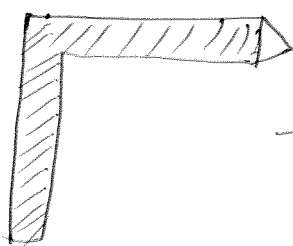


Red balls hanging from a mast

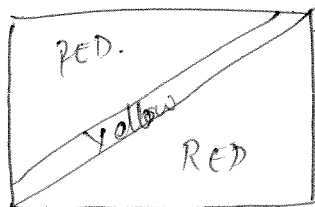
→ Glider flying is in progress.



Green flag → Right hand Circuit pattern in progress.



Red and Yellow striped pointer
→ Right hand Circuit pattern in progress.



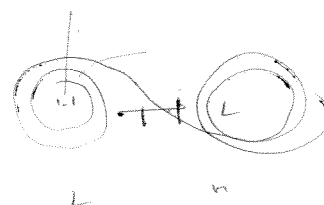
Single Yellow diagonal on Red Background
→ Maneuvering Area is poor. Some Caution.

1. The starboard navigation light of an aircraft is visible over an arc of _____ degrees.

- (a) 70
- (b) 140
- (c) 110*
- (d) 220

2. A prohibited area means an area over which:

- (a) Flight within 2 miles is prohibited
- (b) Flight of aircraft is totally prohibited*
- (c) Low flying is prohibited
- (d) Only instrument flying practice can be done.



3. An aircraft not fitted with navigation lights cannot be flown:

- (a) At any time.
- (b) During hours of night*
- (c) Without permission of aerodrome officer
- (d) Without permission from the DGCA

4. If navigational lights fail at night then

- (a) Land as soon as possible without endangering the safety of the aircraft*
- (b) Fly to destination
- (c) Return to departure aerodrome
- (d) Continue flying

5. Prohibited of flight over Mathura refinery

- (a) With a radius of 1 NM
- (b) With in a radius of 10 km*
- (c) With in a radius of 20 km
- (d) With in a radius of 10 NMs

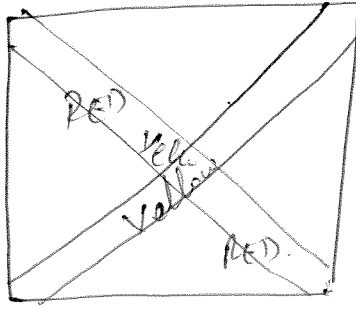
6. Anti-collision light is a light

- (a) Red in colour affixed on top of the control tower
- (b) A red light on the port wing of the aircraft
- (c) A red light visible 30deg above and below the horizon of the aircraft which is visible in all directions*
- (d) The navigation lights of any aircraft are called anti-collision lights

7. An aircraft is following a track of 179° (M) on a VFR plan. The choice of flight levels available to the pilot are

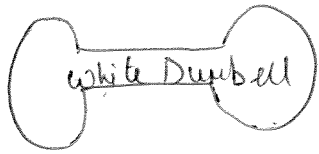
- (a) 50, 70, 90, 150, 170, 190, 210, 230, 250, 270, 290
- (b) 55, 75, 95, 155, 195, 215,, 235, 255, 275, 295,
- (c) 50, 70, 90, 110, 130, 160, 180, 200, 220, 240, 260, 280
- (d) 55, 75, 95, 115, 135,*

No VFR above FL 150.

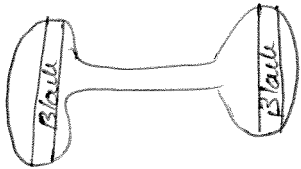


two yellow
diagonal on
red Background

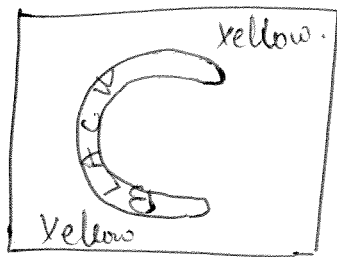
→ Aerodrome is unsafe and
Landing and Take-off is prohibited.



→ movement of planes is Confined to
Hard Paved Surfaces only.



→ Take off and Landing on Hard Surface
But Taxi need not to be Confined to hard Surface only.

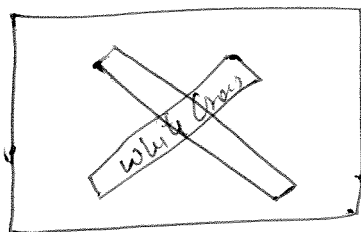


→ Reposting.
ATC office where pilots of visiting A/c should
Report.

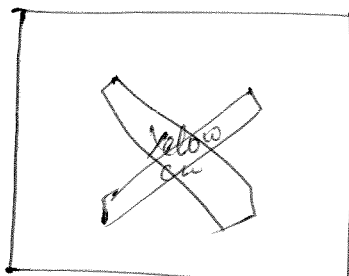
RED	Yellow	Yell
	RED	Yell.
	Y.	RED
		RED

→ Cross hatched
Red-Yellow flag

→ Taxi only in Accordance with
ATC instruction.



→ Runway closed.



→ Taxiway closed.

8. Above flight level 290 the minimum vertical separation between aircraft on reciprocal tracks in RVSM area is

- (a) 1000 feet upto F410*
- (b) 2000 feet upto F410
- (c) 4000 feet upto F460
- (d) 5000 feet upto F460

9. All flight above FL150 flight level are to be cleared IFR irrespective of weather conditions

- (a) F 280
- (b) F 150*
- (c) F 140
- (d) F 100

10. The highest flight level that can be flown in India is

- (a) 460*
- (b) 450
- (c) 440
- (d) 400

11. The most effective ways to use eyes at night is

- (a) Focus only at dim lights At a distance
- (b) Flutter eyes rapidly
- (c) Scan slowly to permit off center viewing *
- (d) Focus on bright lines

12. During flight as a pilot you are subjected to unlawful interference. On SSR you should to set

- (a) Mode A code 7500*
- (b) Mode A code 7600
- (c) Mode A code 7700
- (d) Mode C code 7500

13. A fresh flight plan is to be filed if a scheduled flight is delayed by more than

- (a) One hour
- (b) 30 minutes*
- (c) 15 mins
- (d) 5 mins

14. An airport's rotating beacon operated during daylight hours indicates

- (a) There are obstruction on the airport
- (b) That weather at the airport located in Class D airspace is below basic VFR weather minimum*
- (c) The Air traffic control is not in operation

CPL

Min Age - 18 yrs
Max Age - 65 yrs -

medical validity - 1 year (-40)
medical " - 6 months (+40)

License Validity - 5 years

total flying Hours - 200 Hrs
100 Hrs PIC (solo operator)
20 Hrs X-Country (100 NM)
1 X-Country (300 NM)
5 hours night PIC with 10 hrs
10 landings.

10 Hrs of IR time (min 5 on Actual A/c)

CPL + IR

Min Age - 18 years
Max Age - 65 years

Medical Validity - 1 year (-40)
medical " = 6 months (+40)

CPL License validity - 5 years
IR License validity - 1 year.

total flying Hours - 200 Hrs
100 Hrs PIC (solo operator)
50 Hrs X-Country (100 NM)
1 X-Country (300 NM)
5 Hours night PIC with 10 hrs
10 take-off.

10 Hrs of IR time (min 5 on Actual A/c)

AT PL → airline Transport Pilot License

Min Age - 21 years
Max Age - 65 years.

Medical Validity - 1 year (-40)
" " - 6 months (+40)

Paper: → Radio Nav,

License Validity - 2 years.
total Hrs - 1500 Hrs

PL - 17 yr
Sic CM → 20 years

500 Hrs as PIC (not more than 100 Hrs as Co-pilot)

200 X-Country

100 Hrs night PIC or Co-pilot

100 Hrs of IR (min 50 on Actual A/c)

100 Hrs SIM (not more than 25 hours of Basic Sim)

(d) Runway is blocked

15. During a night flight, you observe a steady red light and a flashing red light ahead and at the same altitude. What is the general direction of movement of the other aircraft?

- (a) The other aircraft is crossing to the left*
- (b) The other aircraft is flying away from you
- (c) The other aircraft is approaching to the right
- (d) The other aircraft is coming head on

16. During a night flight, you observe steady white light and flashing red light ahead and at the same latitude, what is the general direction of movement o the other aircraft.

- (a) The other aircraft is flying away from you*
- (b) The other aircraft is crossing to the Left
- (c) The other aircraft is crossing to the right
- (d) The other aircraft is approaching head-on

17. During the flight, you observe steady green and red lights ahead and at the same altitude, the general direction of movement of the other aircraft?

- (a) The other aircraft is flying away from you
- (b) The other aircraft is flying away from you
- (c) The aircraft is approaching head-on*
- (d) The aircraft is crossing to the right

18. The method of scanning for other aircraft for collision avoidance during the day is to

- (a) Concentrate on the 3-,9-, and 12-o'clock positions
- (b) Regularly spaced eye movement to search each 10 degree sector from left to right*
- (c) Scan small sectors and utilizing off center viewing
- (d) Movements focused at horizon

19. The effective method of scanning for other aircraft for collision avoidance during night the hours is to use

- (a) Concentration on the 3-,9-, and 12-o'clock positions
- (b) Regularly spaced eye movement to search each 10- degree sector
- (c) Scanning small sectors and utilizing off center viewing *
- (d) Objects for 30 secs

20. How can you determine if another aircraft is on a collision course with your aircraft?

- (a) The other aircraft will always appear to get larger and closer at a rapid rate *
- (b) The other of each aircraft is pointed at the same point in space
- (c) There will be no apparent relative motion between your aircraft and the other aircraft
- (d) The nose of other aircraft is pointed towards your aircraft

SAHIL KHURANA
SEC – 7, DWARKA, NEW DELHI
09871866290

21. **Prior to starting each maneuver, pilots should**
- (a) Check altitude, airspeed, and heading
 - (b) Visually scan the entire area for collision avoidance*
 - (c) Announce their intentions to the nearest ATC
 - (d) Fasten seat belt
22. **When activated, an emergency locator transmitter (ELT) transmits on**
- (a) 118.0 and 118.8 MHz
 - (b) 121.5 and 243.0 MHz*
 - (c) 123.0 and 119.0 MHz
 - (d) 118.1 and 123.0 MHz
23. **What is lowest altitude permitted for acrobatic flight?**
- (a) 1000 feet AGL
 - (b) 1500 feet AGL
 - (c) 2000 feet AGL*
 - (d) 6000 feet AGL
24. **No person may operate an aircraft in flight when**
- (a) Flight visibility is less than 5 kms
 - (b) Over any congested area of a city, town below 1000'*
 - (c) Less than 2,500 feet AGL
 - (d) Less than 3,000 AGL
25. **What preparation should a pilot make to adapt the eyes for night flying?**
- (a) Do not stress eyes after sunset until ready for flight
 - (b) Avoid red lights at least 30 minutes before the flight
 - (c) Avoid bright white lights at least 30 minutes before the flight *
 - (d) Avoid any light at least 30 mins before the flight
26. **What is the most effective way to use the eyes during night flight?**
- (a) Look only at far away. dim lights
 - (b) Scan slowly to permit off center viewing*
 - (c) Concentrate directly on each object for a few seconds
 - (d) Scan fast to avoid direct viewing'
27. **Anti-collision lights are required on aircraft above**
- (a) 1500 kg
 - (b) 15000 kg
 - (c) 5700 kgs*
 - (d) All aircraft

London



Australia

Open sky Policies means:- free movement of aeroplanes & Passengers
Not destined for a state in which the
aeroplane has landed.

Cabotage:- Domestic air Service

CONVENTIONS AND FREEDOM

1. ICAO CONVENTION WAS SIGNED IN CHICAGO IN 1944 , 7 TH DEC

2. AS PER THIS CONVENTION FIVE FREEDOMS OF AIR WERE DECLARED

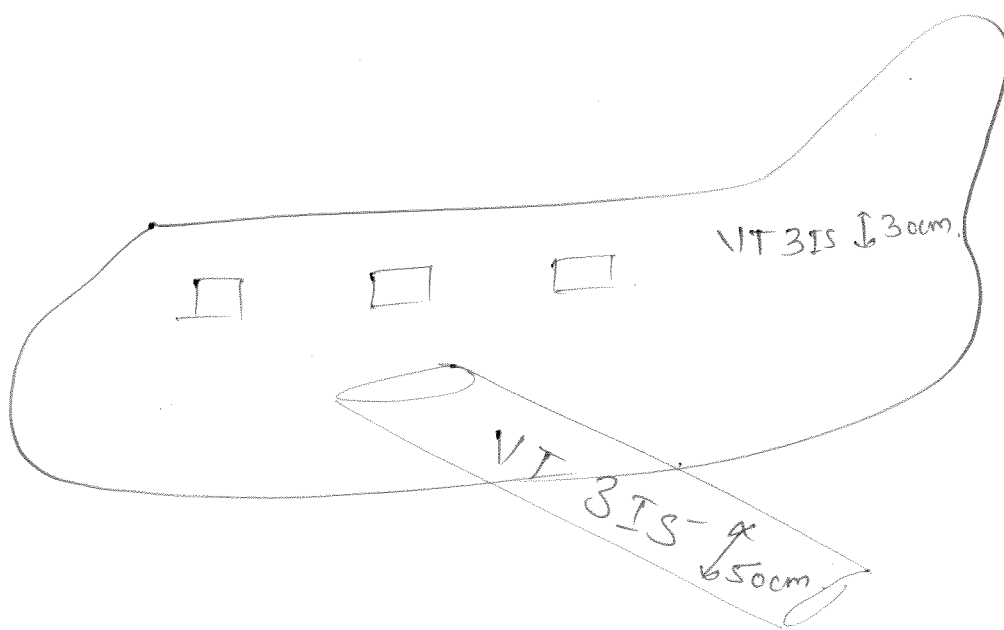
- 3. **FREEDOM 1** – RIGHT TO OVER FLY WITHOUT LANDING
- 4. **FREEDOM 2** – RIGHT TO LAND FOR TECHNICAL PURPOSE (NON TRAFFIC PURPOSE)
- 5. **FREEDOM 3** – RIGHT TO PUT DOWN PASSENGER FROM HOME STATE
- 6. **FREEDOM 4** – RIGHT TO CARRY PASSENGER FOR HOME STATE .
- 7. **FREEDOM 5** – RIGHT TO CARRY PASSENGERS TO THIRD STATE

CONVENTIONS

- 1. TOKYO CONVENTION – OFFENCE ON BOARD AGAINST PENAL LAW
IT IS APPLICABLE TO AIRCRAFT IN THE CONTRACTING STATE AND THE INCIDENT SHOULD IN THE CONTRACTING STATE
- 2. ROME CONVENTION – DEALS WITH DAMAGE CAUSED BY FOREIGN AIRCRAFT
TO THIRD PARTY ON THE SURFACE
- 3. WARSAW CONVENTION – DEALS WITH RESPONSIBILITY FOR CARRIAGE OF PASSENGERS, BAGGAGE AND CARGO.
- 4. MONTERAL CONVENTION – DEALS WITH OFFENCES ON BOARD LEADING TO DAMAGE OF PROPERTY OF AIRCRAFT (UNLAWFUL VIOLENT ACT)
- 5. CHICAGO CONVENTION – 7TH DEC 1944 INDIA IS A SIGNATORY TO CHICAGO CONVENTION
- 6. OBJECTIVE OF ICAO (CHICAGO CONVENTION 1944) – IS TO DEVELOP PRINCIPALS AND TECHNIQUE FOR INTERNATIONAL CIVIL AVIATION
- 7. ICAO ALSO ESTABLISHES S A R P S – STANDARDS AND RECOMMENDED PRACTISES FOR CONTRACTING STATES
- 8. SUBMISSION AND ADOPTING OF SARPS – IS DONE BY AIR NAVIGATION COMMISSION.

~~Cabotage~~ air

Any Contracting state is Denounces the Rule of ICAO it must be informed at least 60 days in Advance notice to Jemo.



~~if your A/c is intercepted.~~

VT 315 is five letter Combination Registration marks

Vt is known as Common mark Assigned by telecommunication union ICAO

315 → is Assigned by state of registry (DGCA)

VT-315 is registration is valid till the A/c is destroyed in Accident.

Registration markings shall be written with no ornamentation.

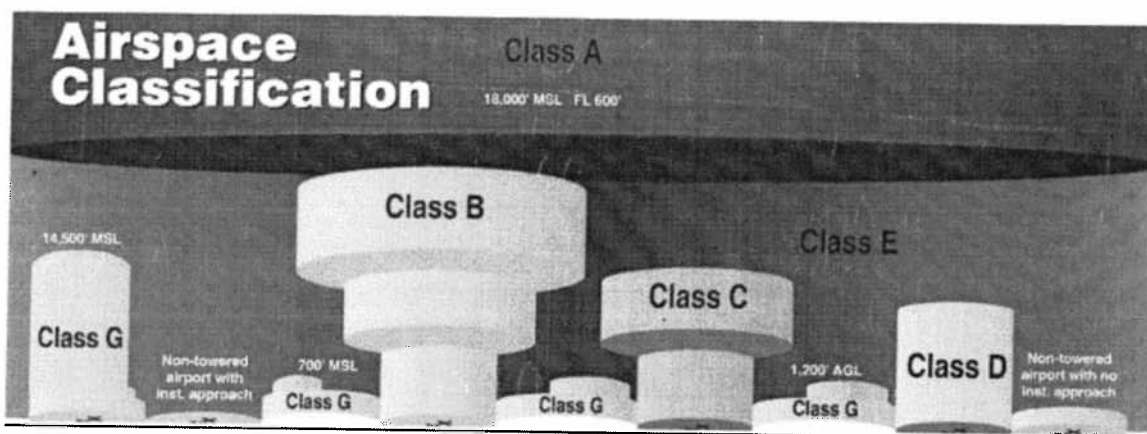
Letters XXX, PAN, TTT shall not be used as Registration markings.

if A/c is intercepted by Another A/c ... Follow the instructions of intercepting A/c and Request clarification from ATC.

Once intercepted by Communicate on ^{CA} 121.5 MHz, 243.0 MHz, 2182 kHz, 306.1 MHz

Abrupt 90° maneuver by the intercepting A/c means you may proceed.

if Communication is not possible on a Common language in world like, make



Air Traffic Control Service A service provided for the purpose of:

a) preventing collisions:

i) between aircraft

ii) on the manoeuvring area between aircraft and obstructions; and

b) expediting and maintaining an orderly flow of air traffic.

Air Traffic Control Unit - A generic term meaning variously, area control centre, approach control office or aerodrome control tower.

Air Traffic Services Airspaces - Airspaces of defined dimensions, alphabetically designated, within which specific types of flights may operate and for which air traffic services and rules of operation are specified.

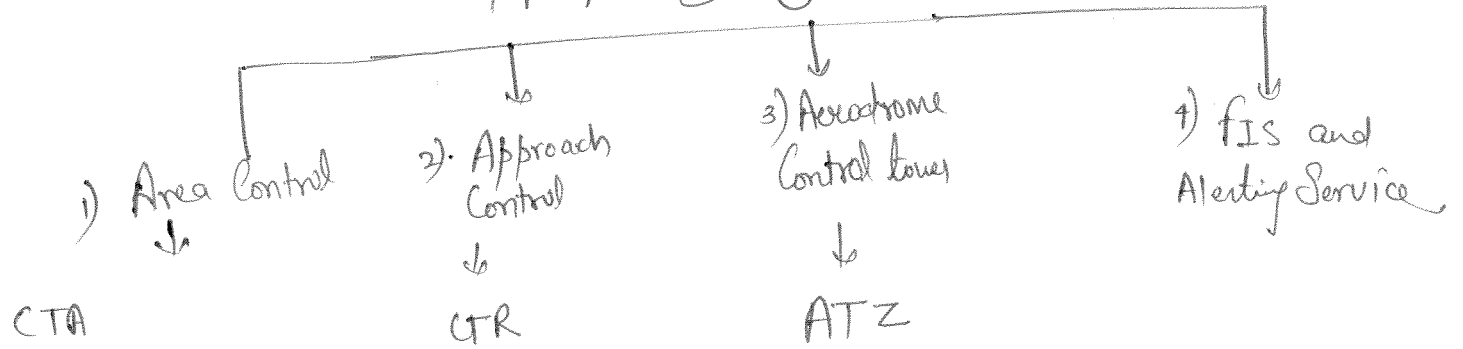
Alerting Service - A service provided to notify appropriate organisations regarding aircraft in need of search and rescue aid, and to assist such organisations as required.

Approach Control Service Air traffic control service for arriving or departing controlled flights.

Controlled Aerodrome An aerodrome at which air traffic control service is provided to aerodrome traffic.

Air Traffic Advisory Service A service provided within advisory airspace to ensure separation, in so far as practical, between aircraft which are operating on IFR flight plans. Only F

ATSU



Control area

CTA [D] 6000ft - FL245

GTR [D] sfc - 6000ft

ATZ - 2000ft

D.



5NM
Safety

700ft AGL

Controlled Airspace *An airspace of defined dimensions within which air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification. Note.- Controlled airspace is a generic term which covers ATS airspace Classes A, B, C, D and E.*

Control Zone *A controlled airspace extending upwards from the surface of the earth to a specified upper limit.*

Control Area *A controlled airspace extending upwards from a specified limit above the earth.*

The lower limit of a CTA is not less than 200 m (700 ft) above MSL or the ground (whichever is higher. Think aerodromes below sea level!).

Control Zone *A controlled airspace extending upwards from the surface of the earth to a specified upper limit.*

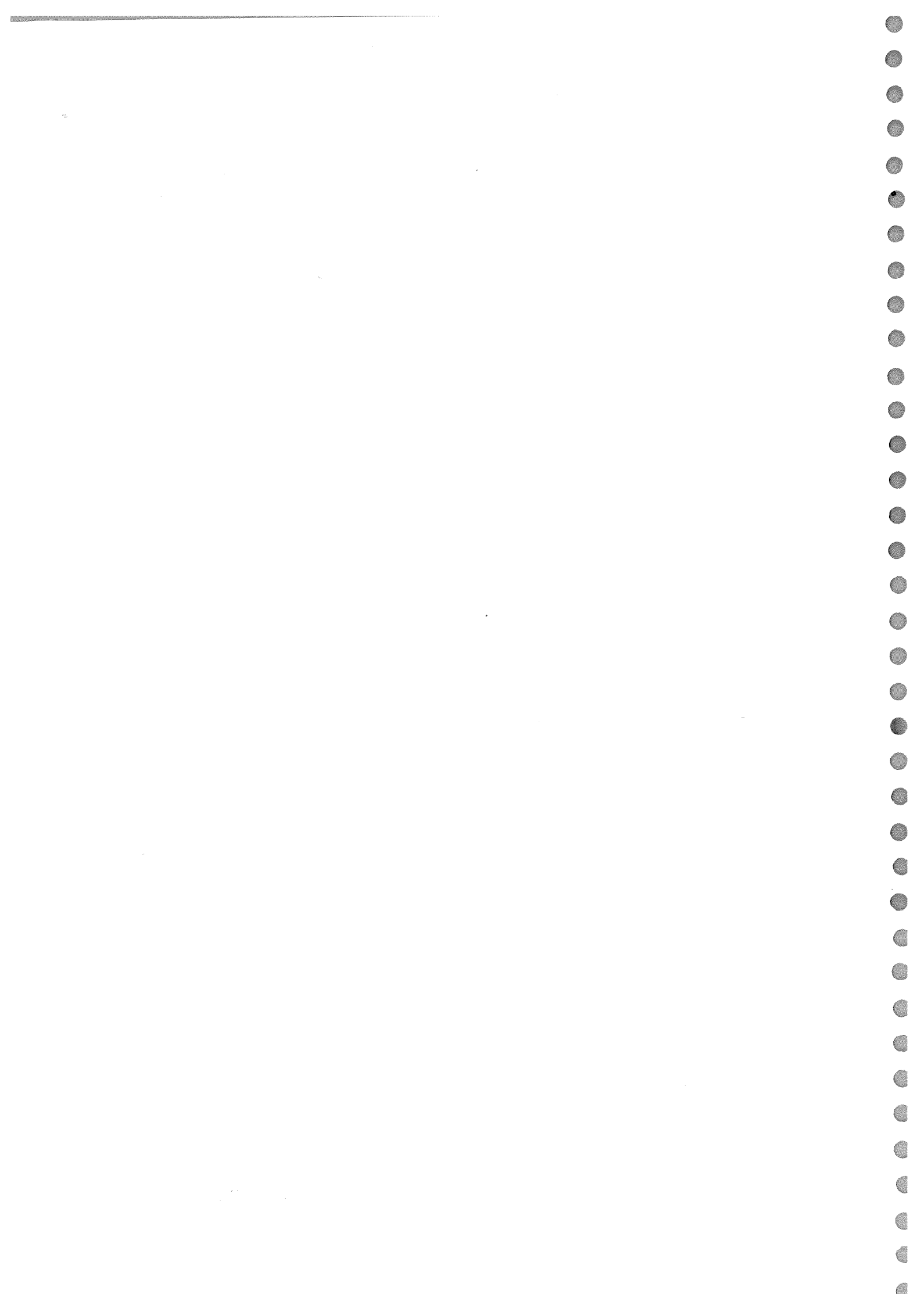
CTRs are defined to include all the airspace, outside of CTAs, used for IFR flights arriving at and departing from aerodromes. The lateral limit of a CTR should extend at least 9.3 km (5 nm) from the centre of the aerodrome, or aerodromes

Class A. *IFR flights only permitted, all flights are subject to ATC and are separated from each other.*

Class B. *IFR and VFR flights are permitted, all flights are subject to ATC and are separated from each other.*

Class C. *IFR and VFR flights are permitted, all flights are subject to ATC, and IFR flights are separated from other IFR flights and from VFR flights. VFR flights are separated from IFR flights and receive information about other VFR flights.*

Class D. *IFR and VFR flights are permitted, all flights are subject to ATC, IFR flights are separated from other IFR flights and receive traffic information concerning VFR flights. VFR flights receive traffic information in respect of all other flights.*



Class E IFR and VFR are permitted, IFR flights are subject to ATC and are separated from other IFR flights. All flights receive traffic information as far as is practicable. Class E is not used for CTRs.

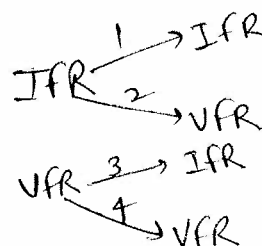
Class F IFR and VFR flights are permitted. All participating IFR flights receive an ATC ADVISORY SERVICE and all flights receive flight information service if requested.

Class G IFR and VFR flights are permitted and receive flight. Information service if requested.

Class	Rules	Service	ATC Clearance	Separation	Traffic Info and avoidance	Speed Limit	Radio Watch	SVFR	Use
+1 A	IFR	ATC	Required	IFR-IFR		No	Yes	Yes	Airway; CTA; CTR; ATZ
+4 B	IFR	ATC	Required	IFR-IFR		No	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC		IFR-VFR VFR-IFR VFR-VFR					
+3 C	IFR	ATC	Required	IFR - IFR		No	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC (FIS)		IFR - VFR VFR - IFR		Yes ✓			
+1 D	IFR	ATC ATC ⁽¹⁾	Required	IFR-IFR		Yes ✓	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC ⁽¹⁾ (FIS)		None	IFR-VFR VFR-IFR VFR-VFR				
+1 (Practical) E	IFR	ATC FIS	Required	IFR-IFR	(Practical) IFR-VFR	Yes ✓	Yes	No	Airway; CTA
	VFR	FIS (FIS)	None	None	VFR-IFR VFR-VFR		No		
+1 (Advisory) F	IFR	ADV ATC	None	IFR-IFR (as far as practicable)		Yes ✓	Yes	No	Advisory Routes
	VFR	(FIS)	None	None			No		
0 G	IFR		None	None		Yes ✓	Yes	No	FIS Routes Open FIR
	VFR	(FIS)					No		

(C-G (includes) FIS Available)
ATC SERVICE SHALL BE PROVIDED TO

- To all IFR flights in A,B,C,D
- To all VFR flights in B,C,D
- To all special VFR flights



Below 10,000ft AMSL
SAHIL KHURANA
may. 250 KTS (Indicated)
SEC-7, DWARKA, NEW DELHI
09871866290

A T I S

ATIS (Automatic Terminal Info Service) $\Rightarrow$ is a Voice Broadcast of Current weather of aerodrome within 5NM and 5000ft

ATIS Broadcast Can be on a Discrete VHF frequency or on a voice channel of VOR, But can never be voice channel of ILS.

ATIS Broadcast should not exceed 30 Sec when practicable

ATIS Broadcast is updated upon receipt of any weather regardless of Content change from Met Department or immediately after a Significant weather change

Preparation and Dissemination of ATIS is Responsibility of air traffic Service.

Movement Area :- Area on the aerodrome including taxiway and Runway only. [Excluding apron]

Movement Area :- The part of aerodrome to be used for take-off, landing and taxing of A/c, consisting of Movement area and the apron(s).

including [apron] Cr. 18, 19, 20

- *All traffic in controlled airspace*

VERTICAL SEPARATION IS PROVIDED

- *Between all flights in A & B airspaces*
- *Between IFR flights in C,D & E*
- *Between IFR flights and VFR flights in C*
- *Between IFR flights and special VFR flight*
- *Between special VFR flights*

ATS ROUTE DESIGNATOR

- *A,B,G,R – Regional network and not RNAV. (International)*
- *L,N,M,P – RNAV with Regional network routes. (International)*
- *Q,T,Y,Z- RNAV not forming part of Regional routes.(Domestic)*
- *H,J,V,W – Not forming Regional network or RNAV routes.(Domestic)*

W is Domestic route in India

U – Upper Airspace

S – Supersonic route

K – Low level routes (helicopters)

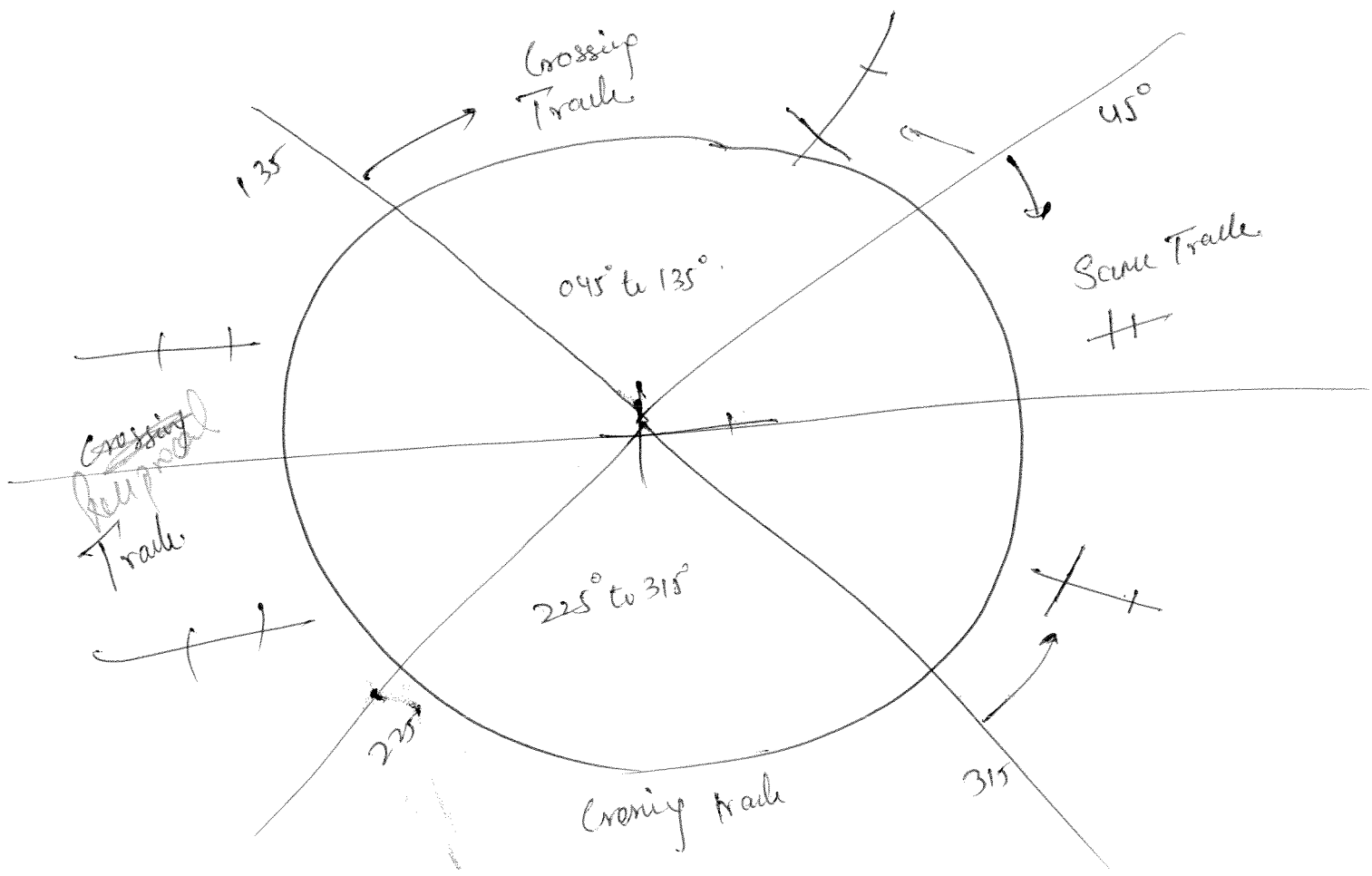
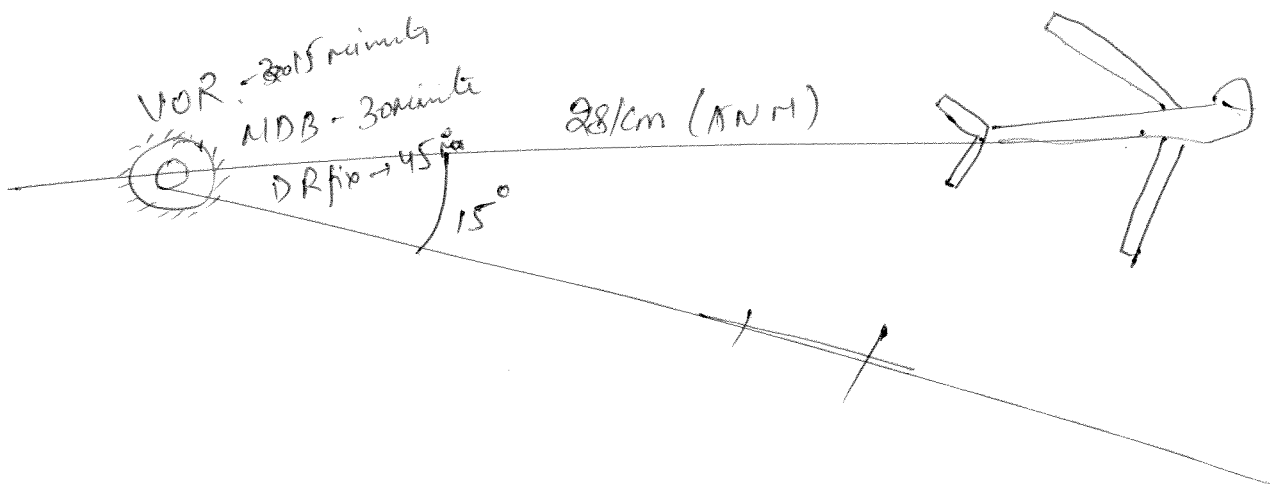
AIRSPACE IN INDIA

DGCA LINE TO REMEMBER

CLASS D: *Airspace in control areas (CTA), terminal areas (TMA), control zones (CTR) and aerodromes traffic zones (ATZ).*

Class E : *Airspace in ATS routes outside CTAs/TMAs/CTRs/ATZs; Except ATS routes segmented listed under class F.*

Class F : *Airspace outside CTAs/TMAs/CTRs/ATZs .*



SEPARATION METHODS AND MINIMA

- HORIZONTAL SEPERATION – By maintaining an interval between A/C operating along the same converging or reciprocal tracks expressed in time or distance.
- LATERAL SEPERATION – By maintaining A/C on Different routes or different geographical locations. [using VOR, NDB, DR fix].

LATERAL SEPERATION USING NAV AIDS

Using the same navigation aid

VOR: at least 15 degrees and at least one aircraft is at a distance of 28 km (15 NM) or more from the facility.

NDB: at least 30 degrees and at a distance of 28 km (15 NM) or more from the facility.

DEAD RECKONING (DR): Diverging by at least 45 degrees and at a distance of 28 (15 NM)

LONGITUDINAL SEPERATION

SAME TRACK - The angular difference of which is less than 45 degrees or more than 315 Degree

RECIPROCAL TRACK- The angular difference of which is more than 135 degrees but less than 225 degrees.

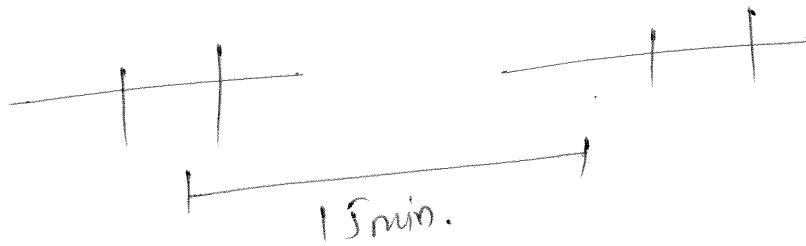
Crossing tracks - Intersecting tracks or portions there of other than those specified in a) and b) above.

→ LONGITUDINAL SEPERATION BASED ON TIME

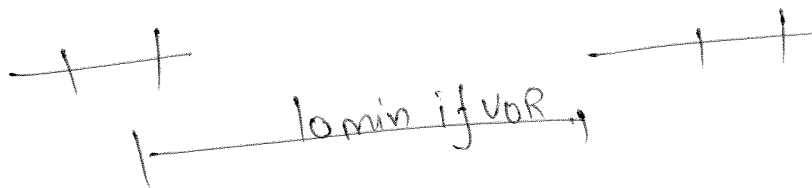
Aircraft flying on the same track:

- ✓ a) 15 minutes
- ✓ b) 10 minutes, if navigation aids permit frequent determination of position and speed

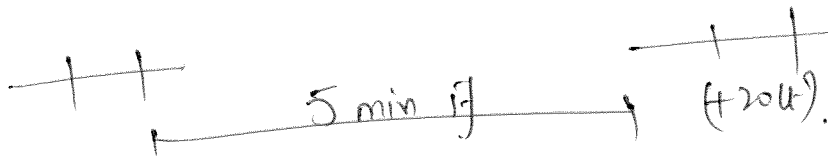
a)



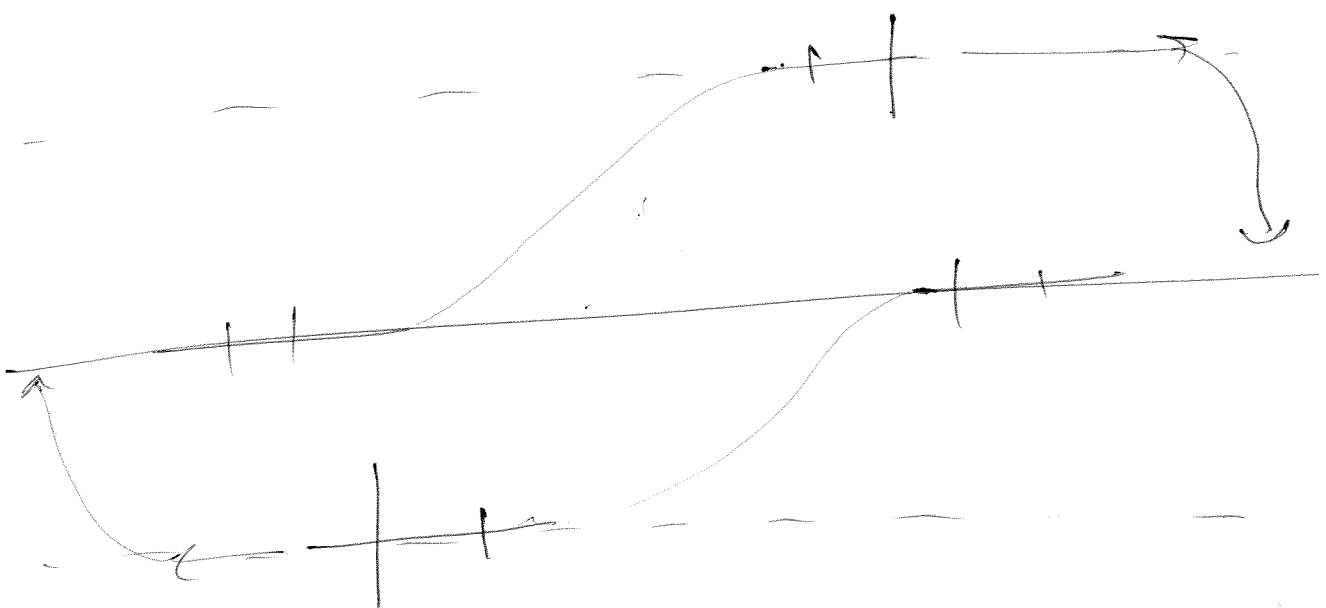
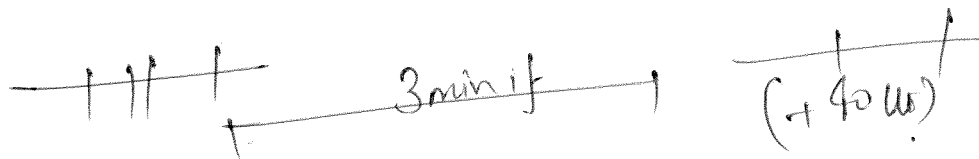
b)



c)



d)



- ✓ c) 5 minutes if the preceding aircraft is maintaining a true airspeed of 37 km/h (20 kt) or more.
- ✓ d) 3 minutes if the preceding aircraft is maintaining a true airspeed of 74 km/h (40 kts) or more.

AIRCRAFT ON CROSSING TRACKS

- ✓ a) 15 minutes
- ✓ b) 10 minutes if navigation aids permit frequent determination of position and speed.

AIRCRAFT CLIMBING OR DESCENDING SAME TRACKS

- a) 15 minutes while vertical separation does not exist
- b) 10 minutes while vertical separation does not exist and only where navigation aids permit frequent determination of position and speed
- ✱ c) 5 minutes while vertical separation does not exist, provided that the level change is commenced within 10 minutes of the time the second aircraft has reported over an exact reporting point.

CLIMBING AND DESCENDING ON CROSSING TRACKS

- a) 15 minutes while vertical separation does not exist
- b) 10 minutes while vertical separation does not exist if navigation aids permit frequent determination of position and speed

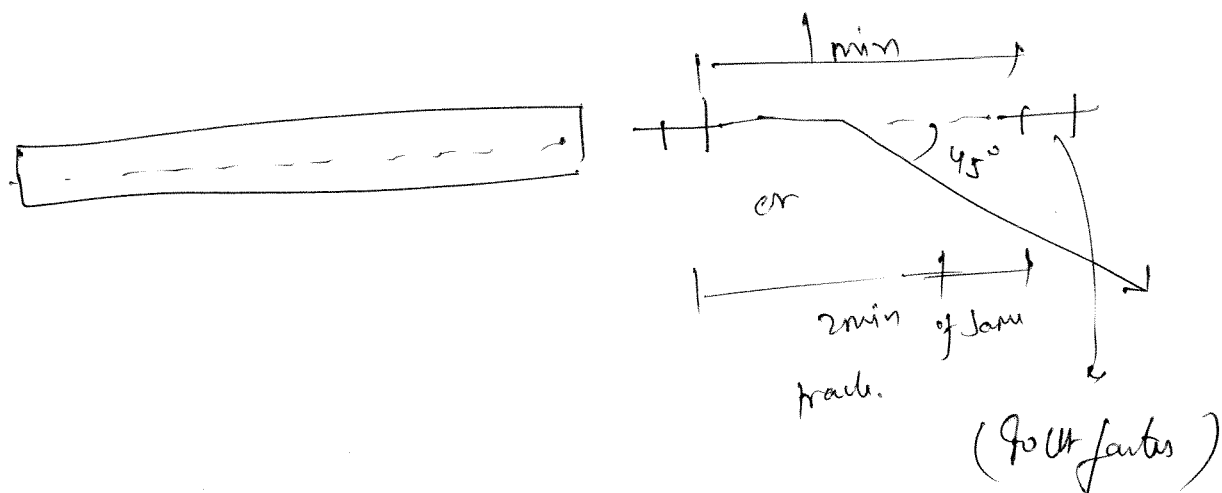
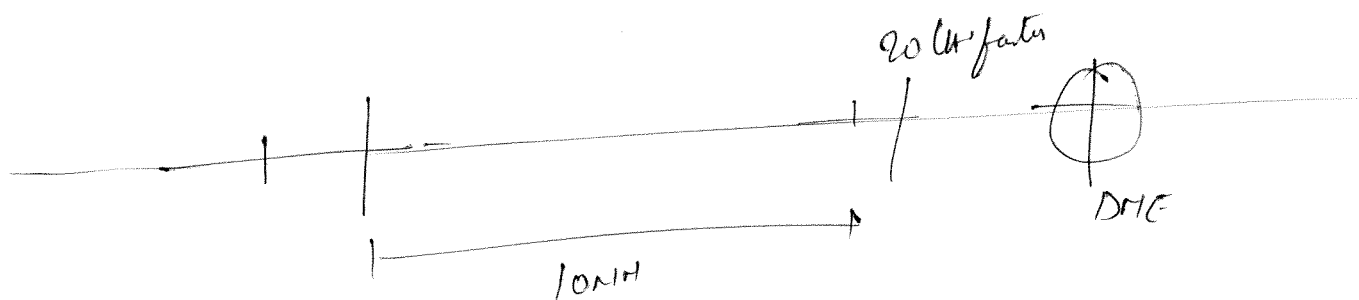
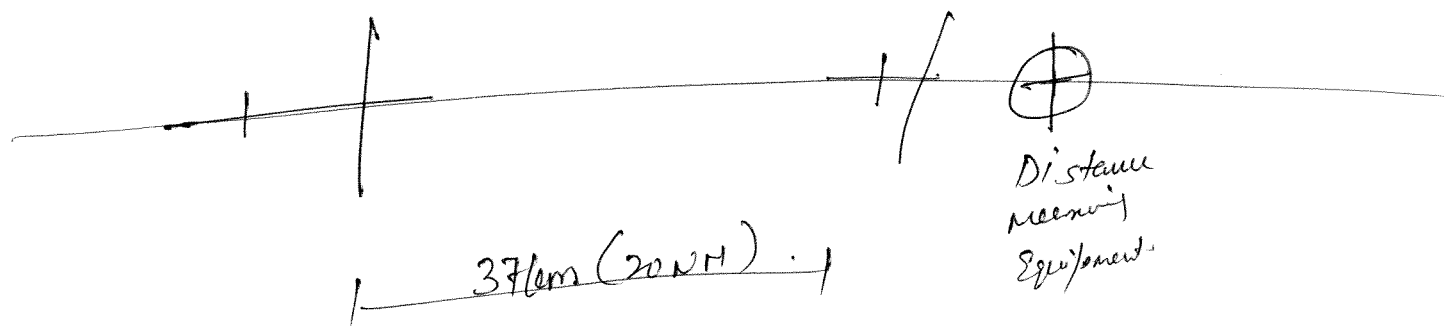
AIRCRAFT ON RECIPROCAL TRACKS

Traffic on reciprocal tracks. Where lateral separation is not provided, vertical separation shall be provided for at least 10 MINUTES prior to and after the time the aircraft are estimated to pass, or are estimated to have passed.

LONGITUDINAL SEPERATION USING DME

Aircraft on the same track:

- ✓ a) 37 km (20 NM), provided each aircraft utilizes "on-track" DME stations



b) 19 km (10 NM) provided the leading aircraft maintains a true airspeed of 37 km/h (20 kt) or more faster than the succeeding aircraft .

Aircraft on the CROSSING tracks:

✓ a) 37 km (20 NM), provided that the relative angle between the tracks is less than 90 degrees.

✓ b) 19 km (10 NM) provided the leading aircraft maintains a true airspeed of 37 km/h (20 kt) or more faster than the succeeding aircraft

AIRCRAFT CLIMBING OR DESCENDING ON SAME TRACKS

Aircraft climbing or descending on the same track:

✓ a) 19 km (10 NM) while vertical separation does not exist, provided each aircraft utilizes "on-track" DME stations.

AIRCRAFT ON RECIPROCAL TRACK

Aircraft on reciprocal tracks - at least 10 NM apart provided that it has been positively established that the aircraft have passed each other.

SEPARATION OF AIRCRAFT HOLDING IN FLIGHT

→ 5 Minutes of flying time from holding area or prescribed distance. of 5 miles .

SEPERATION BETWEEN DEPARTING AIRCRAFT - separation is applied between light or medium a/c taking off behind heavy aircraft. Meaning there is no separation between heavy any heavy

a) 1 minute separation if aircraft are to fly on tracks diverging by at least 45 degrees immediately after take-off so that lateral separation is provided.

b) 2 minutes is required if the tracks are less than 45 degrees apart also if the when the preceding aircraft is 74 km/h (40 kt) or more, faster than the following aircraft and both aircraft propose to follow the same track.

c) Remember this above mentioned is valid when the a/c are taking off from the

1. Same runway

Primary Radar

Only Display A/c Symbol with no Call Sign.

To positively identify an aircraft ATC may ask the A/c to maneuver

Secondary Radar

- Display A/c complete call sign along with pressure Altitude or flight level, speed and mode of Emergency.
- for Positive identification pilot should press ident only when Requested by ATC
- If ident is not working Switch off your transponder to standby and on.
- Example of Secondary Radar is mode C transponder total 4096 codes.
- mode C transponder Has an Accuracy of ± 300 ft

• ATC Has two Radar Approaches.

- 1.) SRA [Secondary Radar Approach] → SRA Approach terminates at 2 miles from Touchdown Zone
- 2.) PAR [Precision Approach Radar] → Terminates at less than $\frac{1}{2}$ miles from Runway.
- 3.) Primary Duty of Radar unit is to provide Radar separation.
- 4.) Horizontal Radar separation shall be 5NM, but on Approach 6NM.
- 5.) minimums Radar separation Ever possible is 3NM.
- 6.) On a Radar Approach ATC (Radar Controller) should never ask an A/c to change his speed in last two of Approach/ govt

2.Parallel runway separated less than 760m.

DEPARTING AIRCRAFT

Leading aircraft	Following Aircraft	Separation Minima	
<u>Heavy</u>	<u>Medium or Light</u>	a. Same runway b. Parallel runway separated by less than 760 m	<u>2 MINUTES</u>
Medium	Light		
Heavy(full length takeoff)	Medium or Light	Departing from a. An intermediate part of the runway or b. Intermediate part of a parallel runway	3 Minutes
Medium (Full length take off)	Light		

SEPERATION BETWEEN ARRIVING AIRCRAFT

LEADING AIRCRAFT	FOLLOWING AIRCRAFT	SEPERATION REQUIRED
Heavy	Medium	2 Minutes
Heavy or Medium	Light	3 Minutes

DISPLACED LANDING THRESHOLD

2 Minutes separation shall be applied between light, medium or heavy aircraft when runways has a displaced threshold.

ALERTING SERVICE

- A/C service provided in need of search and rescue

- 7.] ATC may ask to change speed by ± 20 kts.
- 8.] Radar Contact Means the A/c has been identified on the Radar Display and Radar flight instructions will be provided until Termination.
- 9.] "Resume own navigation" means use your own facility for navigation.
- 10.] Radar Controller shall issue missed Approach instructions if the A/c is not visible on the Radar Display for the last 2 NM of Approach.

OR

- 11.] Landing clearance is not issued by ATC when 2 NM from Touchdown.
- 12.] Delhi Radar Controller must inform Non-Radar Controller (Jaipur) 8 NM Before.
- 13.] 7500 - Hijack, 7600 Comm failure, 770, Distress.
- 14.] Long final means 8 NM from the Approach END.
- 15.] Controlled Aerodrome means it must have a Control tower giving ATC Service.
- 16.] fly Heading 30 Degree means - 030 magnetic Hdg.
- 17.] Departing A/c may Depart in Any Dirⁿ if the Arriving A/c is making a straight in Approach and he is 5 min estimated to be over the Instrument Runway.
- 18.] Controller should inform A/c of mean wind speed change of 10 kts Headwind 5 kts, X-wind change, 2 kts Tailwind change.
- 19.] Radar Controller must Vector see A/c not more than 9.5 NM from other.

- A/C in emergency phase are provided with alerting and search and rescue service by air traffic service unit in following regions (FIRs)

1. VABF 2. VOMF 3. VECF 4. VIDF 5. VEGT

PHASES OF EMERGENCY

1. Uncertainty phase INCERFA

A situation wherein uncertainty exists as to the safety of an aircraft and its occupants

Uncertainty phase when:

- 1) No communication has been received from an aircraft within a period of 30 minutes after the time a communication should have been received.
- 2) An aircraft fails to arrive within 30 minutes of the estimated time of arrival last notified to or estimated by air traffic services units, whichever is the later.

2. Alert phase ALERFA

A situation wherein apprehension exists as to the safety of an aircraft and its occupants.

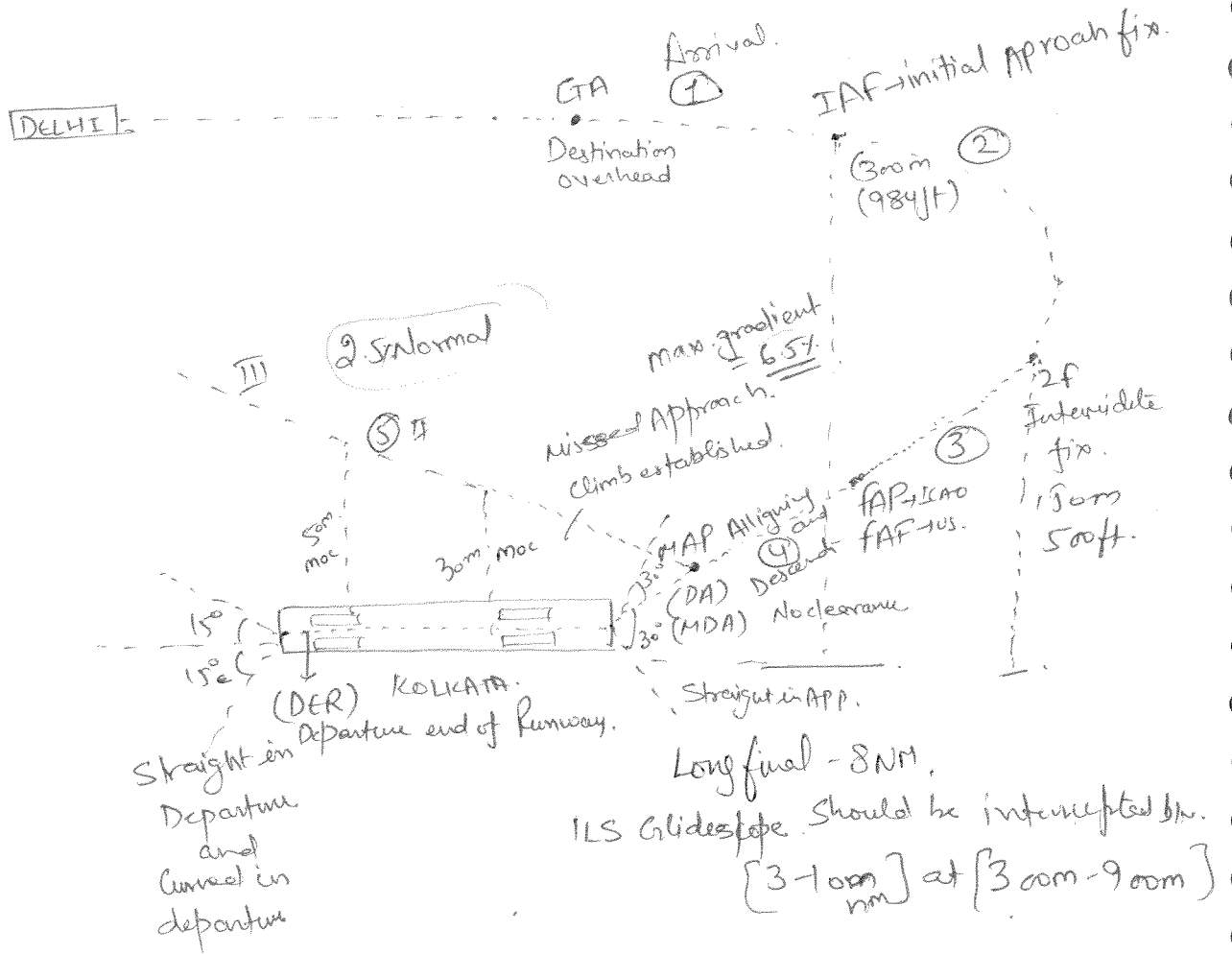
Alert phase when:

- 1) Following the uncertainty phase, subsequent attempts failed to contact it.
- 2) An aircraft has been cleared to land and fails to land within 5 minutes of the estimated time of landing and communication has not been re-established with the aircraft.
- 3) An aircraft is known or believed to be the subject of unlawful interference.

3. Distress phase DETRESFA

[Instrument Approach]

ICAO DOC # 8168.



Timing Departure are given guidance by ATC upto 10km.

ETA → Estimated Time of Arrival (time overhead Destination)

EAT → Expected Approach time → when ATC After Delays will leave the hold for Commencement of Approach.

DER → Departure end of Runway (clearance is '0' ft).

OCA → obstacle clearance Altitude (reference from sea level).

MEHT → Minimum eye height (threshold crossing height).

OIS → obstacle identification System.

MSA → Minimum Sector Altitude (25NM)

ACAS → Airborne collision Avoidance System.

DA(MH) → Decision Altitude (height) → Altitude at MAP in precision Approach

~~MAP~~ → Minimum Descent Altitude → Altitude at MAP in non precision

PA [Precision Approach]. Provides vertical and lateral information (Heading and elevation info).

- ACAS → Airborne Collision Avoiding System → IF promotes situational awareness and keeps searching for conflicting traffic.

29

A situation wherein there is reasonable certainty that an aircraft and its occupants are threatened by grave and imminent danger or require immediate assistance.

Distress phase when:

- 1) **Following the alert phase contact with A/C not established.**
- 2) **The fuel on board is considered to be exhausted.**
- 3) **When aircraft is about to make or has made a forced landing, except when there is reasonable certainty that the aircraft and its occupants are not threatened by grave and imminent danger and do not require immediate assistance.**

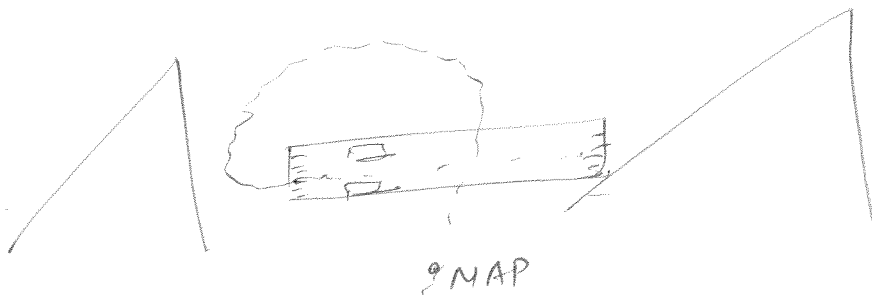
NON-RADAR WAKE TURBULENCE SEPERATION IN

APPROACH CONTROL

<u>TURBULANCE CATEGORY</u>
<u>Heavy 136000 kgs or more (Certified takeoff mass)</u>
<u>Medium – less than 136000 kgs but more than 7000 kg</u>
<u>Light if less than 7000 kgs</u>

RADAR SEPARATION MINIMA

- **5 Nms up to 60 Nms and 6 Nms separation to aircraft in approach and departure phases of flight**
- **10 Nms beyond 60 Nms**



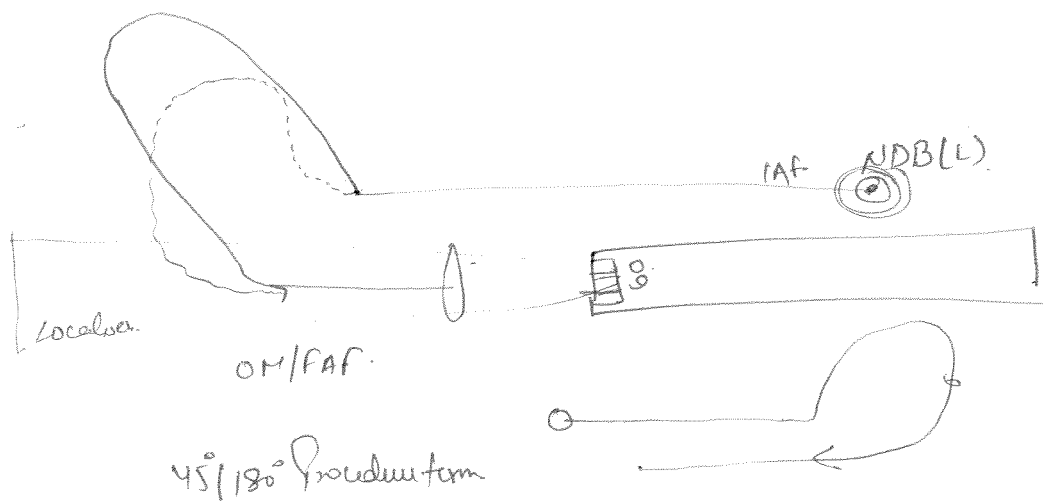
MAP

Circle to Land! - A visual flight maneuver keeping low in sight for landing on runway not suitable for straight in approach.

- You can descend below MDA only when you are in visual contact with the ground and runway light in sight.
- if the Visual Reference is lost.
 - ① Initiate the climb toward the landing runway.
 - ② Initiate a missed approach.

Minimum of instrument approach is decided by the state of Awareness
Ex - Ultralight.

[Procedure Turn]



Procedure Turn

A turn in opposite dirⁿ to permit the Aircraft to proceed along a reciprocal track is a procedure turn.

- Base Turn - At the End of the outBound track and the Beginning of the intermediate for the turn executed is a Base turn.

AERONAUTICAL INFORMATION SERVICES

(AIS) – Annexure 15

Aeronautical Information Publication (AIP)- A publication issued by or with the authority of a State and containing aeronautical information of a lasting character essential to air navigation.

Aeronautical Information Circular (AIC) - A notice containing information that does not qualify for the origination of a NOTAM or for inclusion in the AIP, but which relates to flight safety, air navigation, technical, administrative or legislative matters.

Objectives of the AIS - The objective of the aeronautical information service is to ensure the flow of information necessary for the safety, regularity and efficiency of international air navigation.

CONTENTS - An AIP shall consist of three parts:

Part 1 - General (GEN)

Part 2 - En-route (ENR)

Part 3 - Aerodrome Directory (AD)

An Aeronautical Information Publications is to include General (GEN) the following sections and information: (DGCA)

- ☐ **A statement of the competent authority responsible for the air navigation facilities, services covered by the AIP**
- ☐ **The general conditions under which the services of facilities are available for international use;**
- ☐ **A list of significant differences between the national regulations and practices of the State and the State and the related ICAO Standards, Recommended Practices and Procedures given in a form that would enable a user to differentiate readily between the requirements of the State and the related ICAO provisions**

- ☐ The choice made by a State in each significant case where an alternative course of action is provided for in ICAO Standards, Recommended Practices and Procedures.

AIP Amendment : Permanent changes to the information contained in the AIP.

AIP Supplement : Temporary changes to the information contained in the AIP which are published by means of special pages.

AIRAC : Acronym (Aeronautical Information Regulation and control)
Signifying a system aimed at advance notification, based on common effective dates of circumstances that necessitate significant changes in operating practices Example : Change from NDB to a VOR at any station.

Information concerning the circumstances listed below shall be distributed under the regulated system (AIRAC), basing establishment, withdrawal or significant changes on a series of common effective dates at intervals of 28 days.

The information shall be distributed by the AIS unit at least 42 days in advance of the effective date and the information notified shall not be changed for at least another 28 days after the effective date.

INFORMATION TO BE NOTIFIED BY AIRAC

The establishment, withdrawal of, and premeditated significant changes (including operational trials) to:

Limits (horizontal and vertical), regulations and procedures applicable to:

1. flight information regions;
2. control areas;
3. control zones;

Positions, frequencies, call signs, known irregularities and maintenance periods, of radio navigation aids and communications facilities.

Holding and approach procedures, arrival and departure procedures, noise abatement procedures and any other pertinent A TC procedures.

Meteorological facilities (including broadcasts) and procedures;

Runways and stopways.

NOTICES TO AIRMEN (NOTAM)

Definition: A NOTAM is a notice distributed by means of telecommunications containing information concerning the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.

NOTAMs are originated by the international NOTAM office Mumbai, Kolkata, Delhi and Chennai. The NOTAMs listed for international distribution are promulgated in series A,B,C,D and G as follows :

SERIES A : Contain information in respect of changes/ serviceability of aeronautical facilities, likely to last for more than 2 hours.

SERIES B : Contains information in respect of changes/ unserviceability likely to last for more than 30 minutes but less than 2 hours.

SERIES C : Contain information about domestic flights

SERIES D : Contain information in respect of changes/ unserviceability of aeronautical facilities in respect of Defense- controlled aerodromes only

SERIES G : contain information of general lasting character affecting aircraft operations in general. This series is operated by the International NOTAM office Delhi and issued under DGCA

SNOWTAM - Information concerning snow, ice and standing water on aerodrome pavement areas is to be reported by SNOWTAM.

ASHTAM - Information concerning an operationally significant change in volcanic activity, a volcanic eruption and/or volcanic ash cloud shall, when reported by means of an ASHTAM.

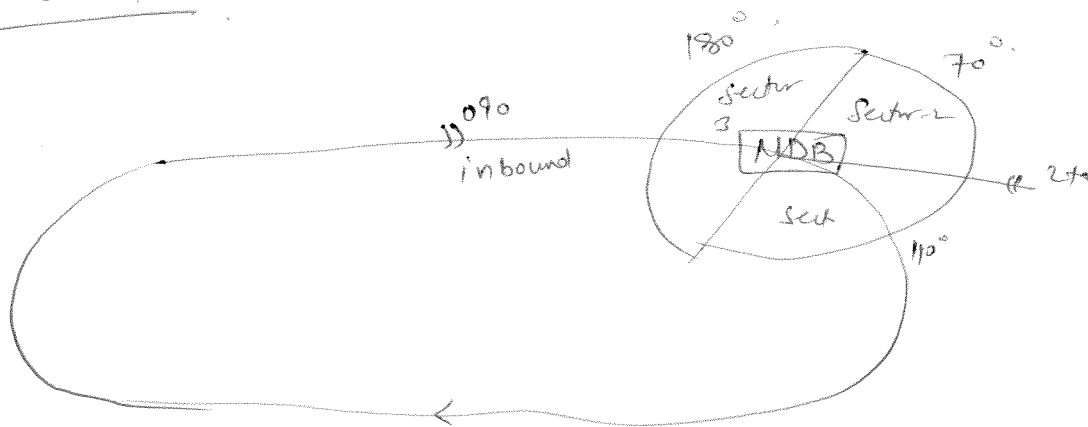
AERONAUTICAL INFORMATION CIRCULAR (AIC) : A notice containing information that does not qualify for the origination of a NOTAM or for inclusion in AIP, but relates to flight safety, air navigation, technical etc..

Example : advice on medical matters of special interest to pilots etc.

ICAO - Annexures

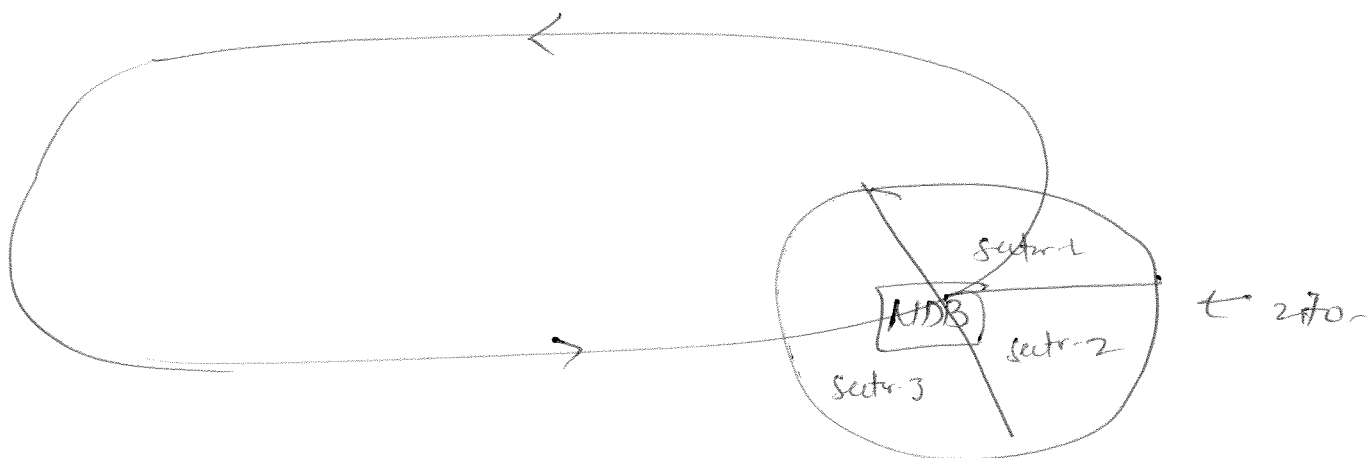
✓ Annex 1	PERSONNEL LICENSING
✓ Annex 2	Rules of the Air
Annex 3	Meteorological Service for Intl Air Navigation
Annex 4	Aeronautical Charts
Annex 5	Units of Measurement Air Ground Ops
Annex 6	Operation of Aircraft
Annex 7	A/C Nationality and Registration Marks
Annex 8	Airworthiness of Aircraft
✓ Annex 9	Facilitation
✓ Annex 10	Aeronautical Telecom
✓ Annex 11	Air Traffic Services
✓ Annex 12	Search and Rescue Air
✓ Annex 13	A/C Accident / Incident Investigation

Standard Hold



- 1) P 1 → Parallel (110°)
- 2) T 2 → Teardrop (70°)
- 3) D 3 → Direct (180°)

Non-standard Hold



- Entry in to the Holding pattern is heading Based.
- All turns in holding pattern are 3/sec or bank angle of 25° which ever is less.
- standard holding pattern turns are made to the right, nonstandard holding pattern turns are made to the left.
- Below 14000 ft 1 min outbound leg.
- Zone of flexibility of the sector boundaries is 5° either side.

<i>Annex 14</i>	<i>Aerodromes</i>
<i>Annex 15</i>	<i>Aeronautical Information Services</i>
<i>Annex 16</i>	<i>Environmental Protection</i>
<i>Annex 17</i>	<i>Security <u>Safeguarding intl. Civil Aviation from Unlawful interference</u></i>
<i>Annex 18</i>	<i>Safe Transport Dangerous Goods by Air</i>

FLIGHT INFORMATION SERVICE (FIS)

A Flight information service (FIS) is to be provided to all aircraft which are likely to be affected by the information and which are:

- ☐ *provided with air traffic control service; or*
- ☐ *otherwise known to the relevant air traffic service units;*

Note: FIS does not relieve the PIC of an aircraft of any responsibilities and he/she has to make the final decision regarding any suggested alteration of flight plan.

Precedence - When ATSU's provide both FIS and ATC service, the provision of ATC service shall have precedence over the provision of FIS whenever the provision of ATC service so requires. In certain circumstances aircraft on final approach, landing, take-off or climb may require essential information without delay, before ATC service information.

Scope of Flight Information Service - FIS includes the provision of pertinent information likely to affect safety and specifically concerning:

- ☐ *SIGMET and AIRMET*
- ☐ *pre-eruption volcanic activity, volcanic eruptions and volcanic ash clouds;*
- ☐ *the release into the atmosphere of radioactive materials or toxic chemicals;*
- ☐ *changes in the serviceability of navigation aids;*
- ☐ *changes in condition of aerodromes and associated facilities, including*
- ☐ *information on the state of the aerodrome movement areas when they are affected by snow, ice or significant depth of water;*
- ☐ *un-manned free balloons*

Voice - ATIS. Voice - automatic terminal information service (Voice-ATIS) broadcasts are provided at aerodromes where there is a requirement to

reduce the communication load on the A TS VHF air-ground communication channels. When provided, Voice-ATIS broadcasts are to comprise:

- a. one broadcast serving arriving aircraft; or*
- b. one broadcast serving departing aircraft; or*
- c. one broadcast serving both arriving and departing aircraft*

ALERTING SERVICE

Provision of Service - It is a requirement for all contracting states of ICAO to provide SAR facilities for all aircraft flying in the airspace of that state. In order to alert the SAR (and other services; fire; ambulance; police; mountain rescue, civil defence), states are required to have a formal system

HOW DOES ATS KNOW THAT THERE IS AN EMERGENCY ?

Uncertainty phase (INCERFA)

Except when no doubt exists as to the safety of the aircraft and its occupants, the uncertainty phase is declared when:

- ☐ *No communication has been received from an aircraft within a period of thirty minutes after the time a communication should have been received, or thirty minutes from the time an unsuccessful attempt to establish communication with the aircraft was first made, whichever is the earlier, or when*
- ☐ *An aircraft fails to arrive within thirty minutes of the estimated time of arrival last notified to, or estimated by ATCUs, whichever is the later*

Alert phase (ALERFA)

When apprehension exists as to the safety of the aircraft and its occupants, or when the distress phase is more appropriate, the alert phase is declared when:

- ☐ *Following the uncertainty phase, subsequent attempts to establish communication with the aircraft have failed to reveal any news of the aircraft;*

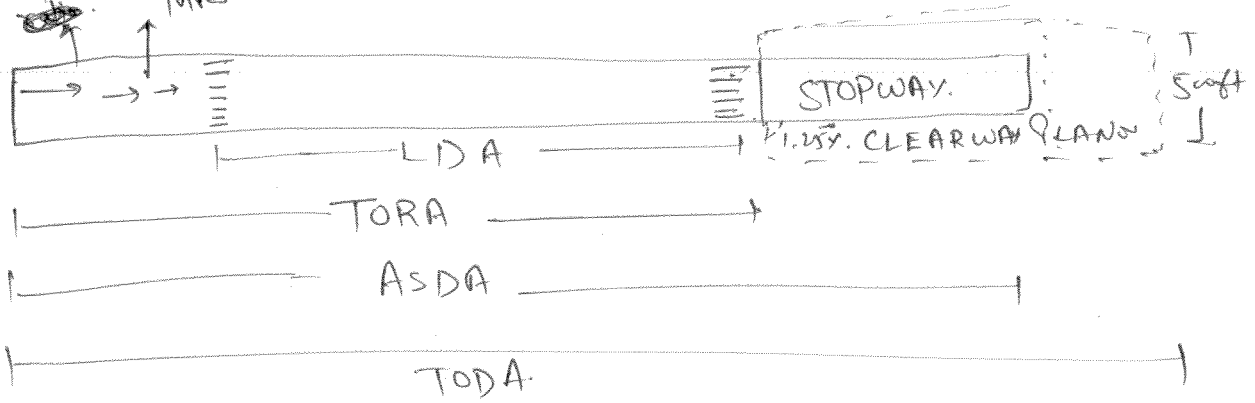
- ☐ *An aircraft has been cleared to land and fails to land within five minutes of the estimated time of landing and communication has not been re-established with the aircraft, or when*
- ☐ *Information has been received which indicates that the operating efficiency of the aircraft has been impaired, but not to the extent that a forced landing is likely, or*
- ☐ *An aircraft is known or believed to be the subject of unlawful interference.*

Distress phase (DETRESFA)

When there is reasonable certainty that the aircraft and its occupants are threatened by grave and imminent danger and do require immediate assistance, the distress phase is declared when:

- ☐ *Following the alert phase further unsuccessful attempts to establish communication with the aircraft point to the probability that the aircraft is in distress, or when*
- ☐ *The fuel on board is considered to be exhausted, or to be insufficient to enable the aircraft to reach safety, or when*
- ☐ *Information is received which indicates that the operating efficiency of the aircraft has been impaired to the extent that a forced landing is likely or when*
- ☐ *Information is received or it is reasonably certain that the aircraft is about to make or has made a forced landing.*

~~Displaced~~ Displaced Threshold → Can only be used for take-off, Not for Landing.



$$\begin{aligned} \text{ASDA} &= \text{TORA} + \text{stopway} \\ \text{TODA} &= \text{TORA} + \text{clearway} \end{aligned}$$

ANNEXURE -14-AERODROMES

Aerodrome - Any area of land or water designed, equipped, set apart or commonly used for affording facilities for the landing and departure of aircraft and includes any area or space, whether on the ground, on the roof of a building or elsewhere, which is designed, equipped or set apart for affording facilities for the landing and departure of aircraft capable of descending or climbing vertically, but shall not include any area the use of which for affording facilities for the landing and departure of aircraft has been abandoned and has not been resumed.

Aerodrome Elevation - The elevation of the highest point of the landing area.

Aerodrome Reference Point - The aerodrome reference point is the geographical location of the aerodrome and the reference centre of its traffic zone where an ATZ is established.

[Gate-17]

APRON - A defined area on a land aerodrome provided for the stationing of aircraft for the embarkation and disembarkation of passengers, the loading and unloading of cargo, and for parking. ^{par.} Ex G-19

MANOEUVRING AREA - That part of an aerodrome provided for the take-off and landing of aircraft and for the movement of aircraft on the surface, excluding the apron and any part of the aerodrome provided for the maintenance of aircraft.

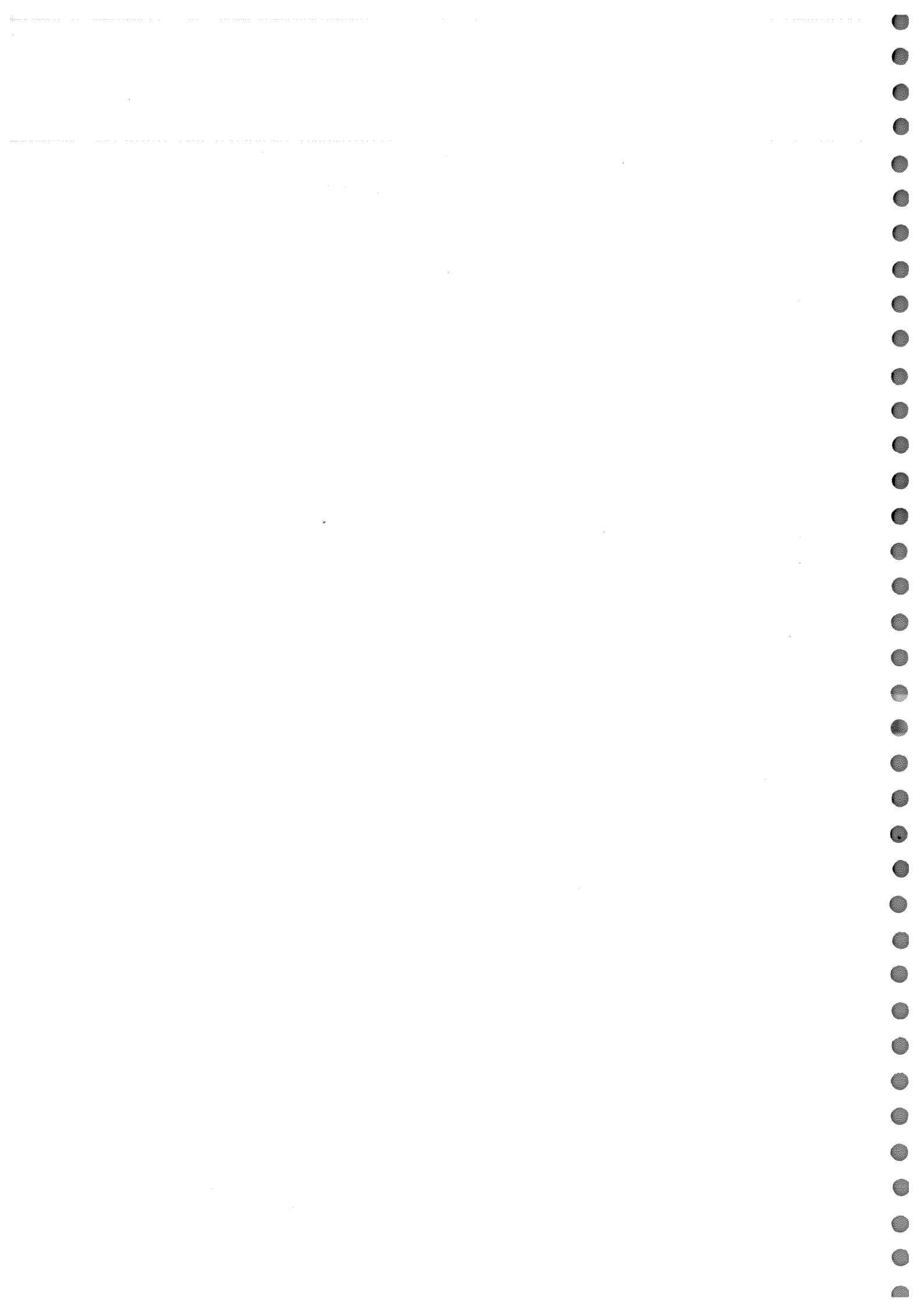
MOVEMENT AREA - That part of an aerodrome intended for the surface movement of aircraft including the manoeuvring area, aprons and any part of the aerodrome provided for the maintenance of aircraft

CLEARWAY - An area at the end of the take-off run available and under the control of the aerodrome licensee, selected or prepared as a suitable area over which an aircraft may make a portion of its initial climb to a specified height(35ft.) ^{SLOPE → not Exceeding 1.25% width → 500ft., length - 1/2 of TORA.}

STOPWAY - A defined rectangular area at the end of the take-off run available, prepared and designated as suitable area in which an aircraft can be stopped in the case of a discontinued take-off.

width same as Runway. no Centreline
Yellow markers opp. to dir.
Shearons.

SAHIL KHURANA
SEC - 7, DWARKA, NEW DELHI
09871866290



DECLARED DISTANCES - The following distances shall be calculated to the nearest metre or foot for a runway intended for the use by international commercial air transport:

- ① ☐ Take-off run available;
- ② ☐ Take-off distance available; = TORA + clearway.
- ☐ Accelerate-stop distance available; and = TORA + stopway
- ☐ Landing distance available.

AERODROME REFERENCE CODE

Use - The reference code, which is used for aerodrome planning purposes, is a simple method of interrelating the specifications concerning the characteristics of aerodromes, so as to provide aerodrome facilities that are suitable for the aeroplanes that are intended to operate at the aerodrome.

Elements - The code is composed of two elements which are related to aeroplane performance characteristics and dimensions.

Element 1 is a number based on the aeroplane reference field length

Element 2 is a letter based on the aeroplane wing span. And outer main gear wheel gear span

Code Element 1		Code Element 2		
Code number	Aeroplane reference field length	Code letter	Wing span	Outer main gear wheel span ⁽¹⁾
1	Less than 800m	A	Up to but not including 15 m	Up to but not including 4.5 m
2	800 m or more, but less than 1 200 m	B	15 m or more, but less than 24 m	4.5 m up to but not including 6 m
3	1 200 m or more, but less than 1 800 m	C	24 m or more, but less than 36 m	6 m up to but not including 9 m
4	1 800 m or more	D	36 m or more, but less than 52 m	9 m up to but not including 14 m
		E	52 m or more, but less than 65 m	9 m up to but not including 14 m
		F	65m up to but not including 80m ⁽²⁾	14 m up to but not including 16 m

Prav

8-

PAVEMENT STRENGTHS - Where pavements are used by aircraft with maximum take off mass greater than 5 700 kg, the strength of the pavement is reported by the aircraft classification number - pavement classification number (ACN-PCN) system.

- ✓ Pavement type for ACN-PCN determination;
- ✓ Subgrade strength category
- ✓ Maximum allowable tire pressure value
- ✓ Evaluation method

➔ The pavement classification number (PCN) reported shall indicate that an aircraft with an aircraft classification number (ACN) equal to or less than the reported PCN can operate on the pavement subject to any limitation on the tire pressure, or aircraft all-up mass for specified aircraft type(s).

Pavement type for ACN-PCN determination: Code

Rigid Pavement	R
Flexible Pavement	F

Subgrade strength category: Code

High strength: characterized by $K = 150$ MN/m³	A
Medium strength: characterized by $K = 80$ MN/m³	B
Low strength: characterized by $K = 40$ MN/m³	C
Ultra low strength: characterized by $K = 20$ MN/m³	D



Maximum allowable tire pressure category:

High: no pressure limit	W
Medium: pressure limited to 1.50 MPa	X
Low: pressure limited to 1.00 MPa	Y
Very low: pressure limited to 0.50 MPa	Z

Technical Evaluation Method

Representing a specific study of the pavement characteristics and application of pavement behaviour technology.	T
Using aircraft experience: representing a knowledge of the specific type and mass of aircraft satisfactorily being supported under regular use.	U

PCN 80 / R / B / W / T → Delhi airport →

ACN → PCN.

If the bearing strength of a rigid pavement, resting on a medium strength subgrade, has been assessed by technical evaluation to be PCN 80 and there is no tire pressure limitation, then the reported information would be

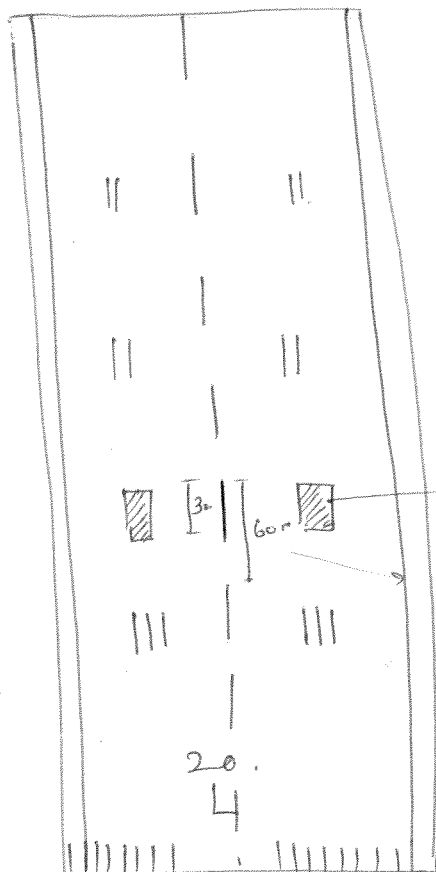
RUNWAYS

USABILITY

For an aerodrome to be used for commercial transport, the schedules agreed must be able to be operated. This means that the runways are not only long enough and strong enough for the aeroplanes to use, but also sited to take into account prevailing meteorological conditions, air traffic requirements, and obstacle clearance. Wind is probably the most important meteorological consideration and runway direction must be based towards the prevailing

TOUCHDOWN ZONE MARK. LDA

- 1 → Less 900m
- 2 → 900-1200m
- 3 → 1200-1500m
- 4 → 1500m-2500m
- 5 → More than 2500m



Threshold.
No of strips.

Run/ WIDTH.

4/2

18m

6/3

23m

8/4

30m

12/5

45m

16/7

60m.

Setback
points

[50-70m]
limits

wind. It is a requirement of ICAO that an aerodrome must be useable for not less than 95% of the notified hours of operation Called as USABILILTY FACTOR

TYPES OF RUNWAY

Runways are described by the types of operations that can be conducted on the runway:

Non Instrument (Visual) Runway

The take-off and landing criteria are determined visually with reference to ground visibility, RVR cloud ceiling, and day/night conditions.

Instrument Runway

A runway to which instrument arrival and departure procedures are applied.

There are two types of instrument runways:

- 1) ☐ Precision runways, → Category 1, Category 2, Category 3.
- 2) ☐ Non-precision runways.

Take-off Runway

A runway used only for take-off operations usually due to excessive obstacles precluding a useful approach.

LENGTH OF RUNWAYS - The actual length of a runway should be adequate to meet the operational requirements of the aeroplanes for which the runway is intended and should be not less than the longest length calculated to correct for local conditions (elevation, temperature, runway slope, humidity and surface characteristics).

WIDTH OF RUNWAYS - The width of a runway should not be less than the dimension specified in the table below.

03L 03R

03L 03C 03R

03L 03R 04L 04R

Code Number	Code Letter					
	A	B	C	D	E	F
1	18 m	18 m	23 m			
2	23 m	23 m	30 m			
3	30 m	30 m	30 m	45 m		
4			45 m	45 m	45 m	60 m

For a precision approach runway the width should not be less than 30 m when the Aerodrome Code Number is 1 or 2.

RUNWAY MARKINGS

RUNWAY DESIGNATOR MARKINGS - A runway designation marking shall consist of a two digit number and on parallel runways shall be supplemented with a letter.

A runway designation marking shall consist of a two digit number and on parallel runways shall be supplemented with a letter.

- ☐ for two parallel runways "L" "R".
- ☐ for three parallel runways "L" "C" "R".
- ☐ for four parallel runways "L" "R" "L" "R" (in this case the QDM for one pair will be increased to differentiate that pair from the other).

Runway centre line marking - A runway centre line marking shall be provided on a paved runway. The centre line marking shall be located along the centre line of the runway between the runway designation markings.

Characteristics - A runway centre line marking shall consist of a line of uniformly spaced stripes and gaps. The length of a stripe plus a gap shall be not less than 50m or more than 75m, The length of each stripe shall be at least equal to the length of the gap or 30m, whichever is greater.

TOUCH DOWN ZONE MARKINGS - Touchdown zone markings consist of pairs of rectangular markings symmetrically placed about the runway centre line with the number of pairs related to the landing distance available. These are 150 m spaced longitudinally



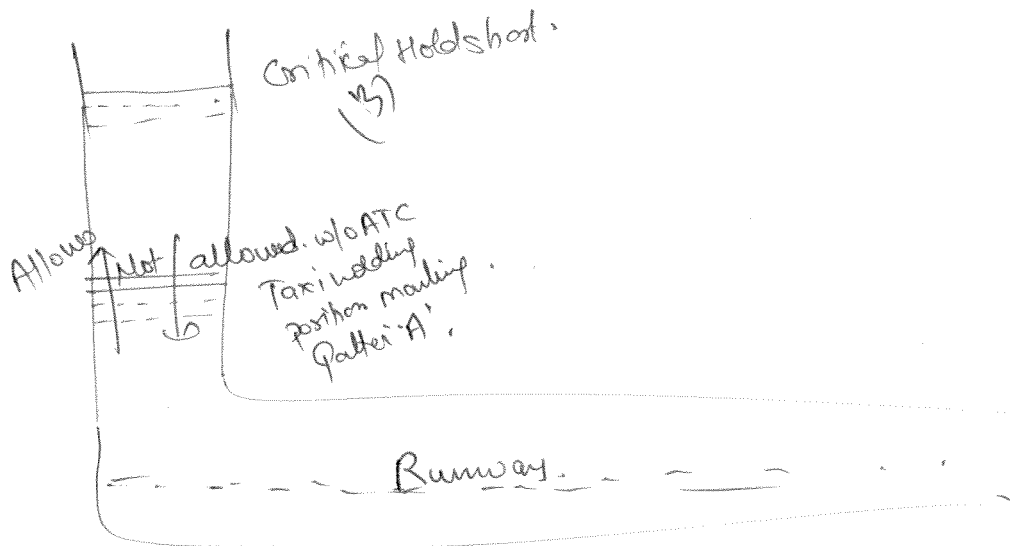
Landing distance available or the distance between thresholds	Pair (s) of markings
less than 900m	1
900m up to but not including 1200m	2
1200m up to but not including 1500m	3
1500 up to but not including 2400m	4
2400m or more	6

The aiming point marking shall commence no closer to the threshold than the distance indicated in the appropriate, except that on a runway equipped with a visual approach slope indicator system, the beginning of the marking shall be coincident with the visual approach slope origin.

Location Of Aiming Point Marking

Location and dimensions	Landing distance available			
	Less than 800m	800m up to but not including 1,200m	1,200m up to but not including 2,400m	2400m and above
Distance from threshold to beginning of marking	150 m	250 m	300 m	400 m

Threshold Marking - A threshold marking is required to be provided at the thresholds of paved instrument runways, and of paved non-instrument runways where the code number is 3 or 4 and the runway is intended for use by international commercial air transport (CAT)



① A/c should hold short in such a way so that no part of aircraft extend beyond the hold-short line.

② A/c Cannot cross solid dash line without permission from ~~TA~~ ATC.

③ Where ~~to~~ Hold-short markings are not painted. A/c should stop 50m from Runway.

Characteristics - A runway threshold marking is a pattern of longitudinal stripes of uniform dimensions disposed symmetrically about the centre line of the runway

TAXIWAY MARKINGS.

- ☐ Taxiway markings and aircraft stand markings are yellow.
- ☐ Taxiway centre line marking shall be provided on a paved runway when the runway is part of a standard taxi-route and:
 - I. There is no runway centre line marking; or
 - II. Where the taxiway centre line is not coincident with the runway centre line.

RUNWAY HOLDING POSITION MARKING

A runway holding position marking is placed at runway holding position. The holding point itself is indicated by the mandatory sign(s) at the side of the taxiway. There are two types of runway holding marking: pattern A and pattern B.

AERODROME LIGHTING -An aerodrome at night is a profusion of lights. Apart from the main stadium lighting around the apron and passenger areas, the runways are lit as are the taxiways; buildings and vehicles have lights on them as do aircraft. All the coloured lights have a purpose and the arrangements of lights also have meanings.

VARIABLE INTENSITY LIGHTS

The intensity (brilliance) of the following lights is to be variable:

- Approach lighting systems [Simple, Category 1, Category 2, 3]
- Runway edge lights white, last 600m yellow.
- Runway threshold lights → Unidirectional, steady green light.
- Runway end lights → Red.

Last 900m

Last 300 46

- Runway centre line lights *white, alternate red & white, red.*
- Runway touchdown zone lights *white.*

AERODROME BEACON

- ☐ **Flashing white and green for civilian land airports.** *Positive flashing green only.*
- ☐ **Flashing white and yellow for a water airport.**
- ☐ **Flashing white, yellow, and green, for a heliport.**
- ☐ **Two quick, white flashes followed by a green flash identifies a military airport.**

An aerodrome beacon is usually situated on top of the control tower or at an elevated position elsewhere on the aerodrome,

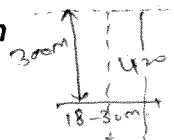
APPROACH LIGHTING SYSTEMS

Approach lighting systems are provided to enhance the ability of a pilot to visually acquire the runway at night or in low visibility. For an instrument runway the approach lighting is an essential part of the instrument system and can vary in complexity from a simple system consisting of a centre line and a cross bar.

- 1] ☐ **SIMPLE APPROACH LIGHTING SYSTEM**
- 2] ☐ **PRECISION APPROACH CAT I LIGHTING SYSTEM**
- 3] ☐ **PRECISION APPROACH CATEGORY II AND III LIGHTING SYSTEM**

SIMPLE APPROACH LIGHTING SYSTEM

- ☐ **The system consists of a single light source centre line extending from the threshold of the runway along the approach path for 420 m.**
- ☐ **A single crossbar min 18 m wide and max 30 m wide is placed at a point 300 m from the threshold.**
- ☐ **The lights are fixed (not flashing) variable intensity white, showing towards an approaching aircraft.**

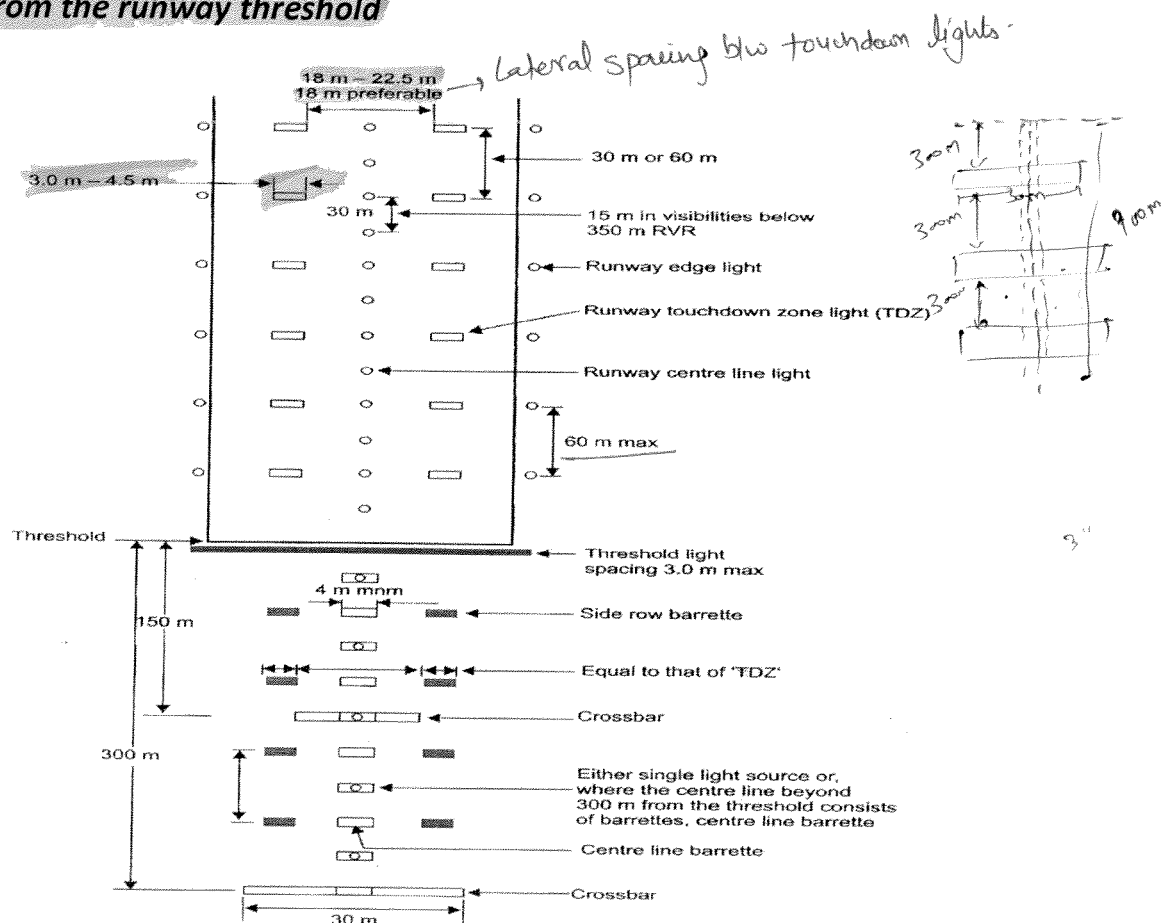


- ☐ The centreline may be made up of barrettes not more than 3 m wide.
- ☐ The system may be used for a non precision instrument runway.

BARRETTES

ICAO specifies standards for approach lighting based on the use of barrettes (or small bars). A barrette consists of a line of 4 or 5 closely spaced lights forming a small bar.

- ☐ A precision approach category I lighting system shall consist of a row of lights on the extended centre line of the runway extending, wherever possible, over a distance of 900m from the runway threshold with a row of lights forming a crossbar 30m in length at a distance of 300m from the runway threshold



- ☐ The centreline must be exactly 900 m long.
- ☐ The spacing between the crossbars is 150 m.

W W R R.
 ● ● ● ●
 well Above glide path.

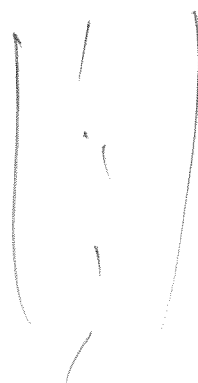
● ● ● ●
 slight above glide path.

● ● ● ●
 on or near glide path.

● ● ● ●
 below glide path,

All fed.
 ● ● ● ●
 well below glide path

one white one fed.
 ○ ○



- ☐ The lights forming the centre line shall be placed at longitudinal intervals of 30m with the innermost lights located 30m from the threshold.
- ☐ The lights forming the side rows shall be placed on each side of the centre line, at a longitudinal spacing equal to that of the centre line lights and with the first light located 30m from the threshold.
- ☐ The lateral spacing (or gauge) between the innermost lights of the side row shall be not less than 18m nor more than 22.5m

PRECISION APPROACH PATH INDICATOR (PAPI)

PAPI are the latest generation of visual approach slope indicators which are used to give the pilot a visual indication of the vertical deviation from the nominal glide path.

- ✓ When on or close to the approach slope, see the two units nearest the runway as red and the two units farthest from the runway as white;
- ✓ When above the approach slope, see the one unit nearest the runway as red and the three units farthest from the runway as white; and when further above the Approach slope, see all the units as white;
- ✓ When below the approach slope, see the three units nearest the runway as red and the unit farthest from the runway as white; and when further below the approach slope, see all the units as red.

APAPI

The APAPI uses only two light sources and gives indication:

- ☐ Two white - above the glide path
- ☐ One white one red - on the glide path
- ☐ Two red - below the glide path

RUNWAY LIGHTS

RUNWAY EDGE LIGHTS

- ☐ Runway Edge lights are used on runways used at night or in low visibility operations (RVR less than 800 m).
- ☐ They are fixed (not flashing), white, variable in intensity and unidirectional (facing the approach direction)
- ☐ For a CAT I runway, the last 600 m or 1/3 of the runway length, the edge lights are possibly yellow, indicating the proximity of the end of the runway. This is called a caution zone.

RUNWAY THRESHOLD AND WING BARS

Threshold lights are provided for runways with edge lights and additional wing bars where there is a displaced threshold. They are fixed, green, variable intensity, and unidirectional facing up the approach.

RUNWAY END LIGHTS

- ☐ Runway end lights shall be fixed unidirectional lights showing red in the direction of the runway.
- ☒ Where a stopway is created at the end of the runway all four sides of the stopway are outlined in red lights.

RUNWAY CENTRELINE LIGHTS

- ☐ Centreline lights are required on CAT II/III runways.
- ☐ They are fixed, white, and variable in intensity, and are unidirectional showing towards the approach end of the runway.
- ☐ Runway centre line lights shall be fixed lights showing variable white from the threshold to the point 900m from the runway end; alternate

red and variable white from 900m to 300m from the runway end; and red from 300m to the runway end, except that:

for runways less than 1,800m in length, the alternate red and variable white lights shall extend from the mid-point of the runway usable for landing to 300m from the runway end.

RUNWAY TOUCHDOWN ZONE LIGHTS

- ☐ *Touchdown zone lights shall be provided in the touchdown zone of a precision approach runway category II or III.*
- ☐ *Touchdown zone lights shall extend from the threshold for a longitudinal distance of 900m, except that, on runways less than 1,800m in length, the system shall be shortened so that it does not extend beyond the midpoint of the runway*
- ☐ *The longitudinal spacing between pairs of barrettes shall be either 30m or 60m. A barrette shall be composed of at least three lights with a spacing between the lights of not more than 1.5m.*
- ☐ *Touchdown zone lights shall be fixed unidirectional lights showing variable white.*

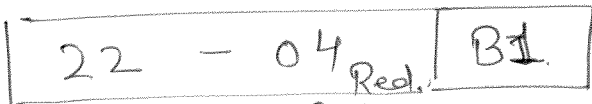
STOPWAY LIGHTS

Stopway lights shall be fixed unidirectional lights showing red in the direction of the runway.

TAXIWAY LIGHTING

- ☐ *Taxiway Lighting system consists of centreline lights, edge lights, guard lights, and stop lights at holding points.*
- ☐ *TAXIWAY EDGE LIGHTS -The lights are to be of fixed variable intensity showing BLUE, showing all round and up to 30° above the horizontal.*

Aerodrome Signs.



Background

- Red Background.
- white letters.

Location

BB.

[Black Background]
yellow letters
shows your location

Direction

Yellow Background
Black letters

Destination

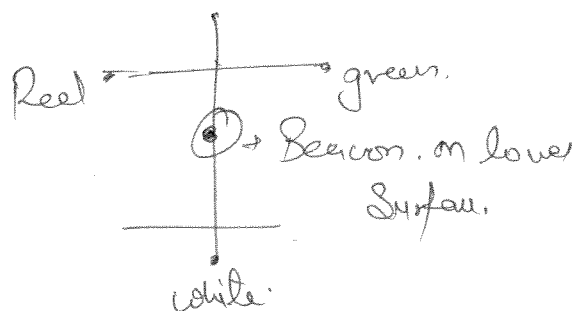


Yellow letters

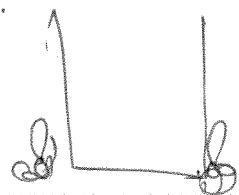
for obstacle above 150m within Radius 15Km. from ARP.
are lit by high Intensity flashing white by day and low intensity steady red by night.

All the vehicle on aerodrome should have flashing Yellow or Orange Beacons.

Leading lights flashing white → They lead you towards runway.



Runway end identification lights are flashing white.



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TAXIWAY CENTRE LINE LIGHTS -

- ☐ Taxiway centre line lights on a taxiway shall be fixed light showing green such that the light is visible only from the aeroplanes on or in the vicinity of the taxiway.
- ☐ Taxiway centre line lights on an EXIT TAXIWAY shall be fixed lights. Alternate lights shall show green and yellow from their beginning near the runway centre line and thereafter all lights shall show green

MARSHALL SIGNALS

TURN TO YOUR LEFT

Right arm downward, left arm repeatedly moved upward backward.

Speed of arm movement indicating rate of turn.

TURN TO YOUR RIGHT

Left arm downward, right arm repeatedly moved upward backward

Speed of arm movement indicating rate of turn.

MOVE AHEAD

Arms a little aside, palms facing backward and repeatedly moved upward-backward from shoulder height.

CHOCKS INSERTED

Arms down, palms facing inwards, move arms from extended position inwards.

CHOCKS REMOVED

Arms down, palms facing outwards, move arms outwards.

START ENGINE(S)

Left hand overhead with appropriate number of fingers extended, to indicate the number of the engine to be started, and circular motion of right hand at head level.

CUT ENGINES

Either arm and hand level with shoulder, hand across the throat, palm downward. The hand is moved sideways with the arm remaining bent.

SLOW DOWN

Arms down with palms toward ground.

STOP

Arms repeatedly crossed above the head (the rapidity of the arm movement should be related to the urgency of the stop ie the faster the movement the quicker the stop).

Symptoms of Hypoxia.

→ Personality change.

• Asense of well being [Euphoria] ★

★ → Impaired Judgement.

[Euphoria.]

[EUPHORIA]

→ Headach

→ Tingling in hand & feet

→ Hyperventilation

★ → muscular impairment
→ memory Impairment
→ Sensory loss

★ → Tunnel vision or by 30° cone

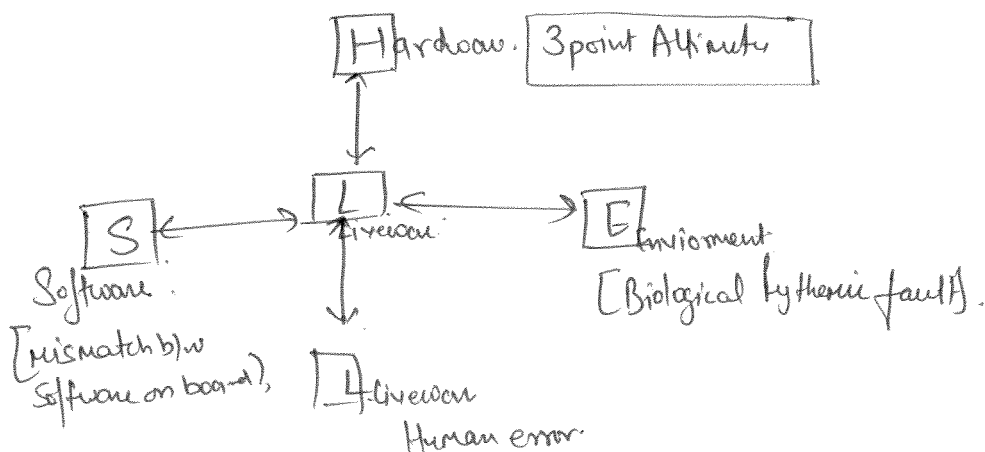
→ Impairment of Consciousness

★ → Cyanosis (a bluing in blood extremities).

→ Formication (A feeling of ants under the skin).

★ → Unconsciousness and ultimately death.

SHELL Model



8-12 marks In CPL.

7 Questions
14 Marks at least
Very Important.

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HUMAN FACTORS

HYPOXIA is a state of oxygen deficiency in the body sufficient to impair functions of the brain and other organs

☐ **ANAEMIC HYPOXIA**

Anaemic Hypoxia is caused by the inability of the blood to carry oxygen and may be due to a medical condition (anaemia) or to carbon monoxide poisoning

☐ **HYPOXIC HYPOXIA.**

The term for the effects of a shortage of oxygen is Hypoxic Hypoxia. This can result from a number of reasons. But the most important reason, as far as pilots are concerned, is altitude.

PREVENTION OF HYPOXIA

For protection, Pilots are encouraged to use supplemental oxygen above 10,000 during the Day, and 5000' above at night.

DALTONS LAW

1. DALTON LAW EXPLAINS ALTITUDE HYPOXIA, HE EXPLAINED THAT THE TOTAL PRESSURE IS EQUAL TO SUM OF PARTIAL PRESSURE OF GASSES
2. CHEMICAL COMPOSITION IS 78 % N₂, 21 % O₂, .9 % ARGON, .03 % CO₂, REST RARE GASSES.
3. OXYGEN COMBINED WITH HB IN BLOOD IS TRANSPORTED BY RBC (RED BLOOD CELLS)
4. PERCENTAGE OF O₂ IS SAME AT ALL LEVELS - 21 %
5. BAROMETRIC PRESSURE AT 18000 FT IS HALF
6. YOU CAN SURVIVE AT ANY ALTITUDE IF ENOUGH O₂ AND HEAT IS PROVIDED
7. BREATHING 100 % O₂ AT HIGH ALTITUDE (38000) IS EQUIVALENT TO BREATHING AMBIENT AIR AT 10000 FT.
8. BREATHING 100 % O₂ WITHOUT PRESSURE AT 38000 FT WILL CAUSE HYPOXIA.

HYPERVENTILATION

IT IS ANOTHER WORD FOR **OVERBREATHING**, LUNGS VENTILATION IN EXCESS OF BODY NEEDS (**LACK OF CO₂ IN THE BODY**)

CAUSES OF HYPERVENTILATION

- **ANXIETY**
- **SHOCK**
- **HEAT**
- **G FORCES**
- **MOTION SICKNESS**

SYMPTOMS OF HYPERVENTILATION

ANXIETY

FEAR

DISTRESS

SPASMS → Expansion and Contraction of muscles.

1. **INCREASED HEART RATE**

3. Dizziness

2. **LOSS OF CONSCIOUSNESS**

+ CO₂ ---- BLOOD ACIDIC

- CO₂ ---- BLOOD ALKALINE

Treatment of Hyperventilation

The classic way to treat a patient suffering from Hyperventilation is to **make him/her breathe into a paper bag.**

HYPOXIA or HYPERVENTILATION?

The natural reaction to a shortage of oxygen is for the body to **try to obtain more air by breathing faster and deeper.** The hypoxic individual may Hyperventilate in an effort to get more oxygen, but this is of little value when in an environment of low ambient pressure.

In night it can be difficult to distinguish the symptoms of Hypoxia and Hyperventilation. The appropriate response of pilots must be to assume the worst

DO NOT ASSUME HYPERVENTILATION IF IT COULD BE HYPOXIA

HYPERVENTILATION - AFTER UNCONSCIOUSNESS – RECOVERY

HYPOXIA - AFTER UNCONSCIOUSNESS – DEATH

CO is an colourless, odourless and tasteless gas contained in exhaust fumes. It can significantly reduce the ability of the blood to carry oxygen and leads to HYPOXIA.

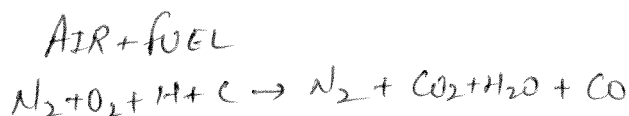
- CARBON MONOXIDE POISONING

✶ **CO POISONING IS LIKELY TO OCCUR IN AEROPLANES WHERE CABIN HEAT IS SUPPLIED FROM THE EXHAUST COATING**

✶ **HB HAS MORE AFFINITY TOWARDS CO .**

SYMPTOMS OF CO –

1. **HEADACHE**
2. **DIZZINESS**
3. **NAUSEA**
4. **IMPAIRED VISION**



ACTION TO BE TAKEN

SWITCH OFF CABIN HEAT , OPEN CABIN VENTILATION , USE FRESH AIR , USE O2 IF AVIALBLE

LAND AS SOON AS POSSIBLE

CABIN DECOMPRESSION

Loss of cabin pressurisation can occur in flight.

Decompression sickness

✶ **DUE TO CABIN PRESSURE LOSS WHEN FLYING AT HIGHER ALTITUDES (ABOVE 18000 FT)**

Creep → Bubble moving under the skin (N₂)
Chokes → Lungs gets blocked due to N₂ big bubble.
Bends → N₂ Bubble stuck in the knees.

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HENRY EXPLAINED THE RELEASE OF NITROGEN GAS BUBBLES IN THE BODY FLUIDS CAUSING BENDS, CHOKES, CREEPS, NEUROLOGICAL SYMPTOMS

PAX SUFFERING FROM RAPID DECOMPRESSION SHOULD WAIT FOR 12 HOURS BEFORE NEXT FLIGHT

SYMPTOMS OF DECOMPRESSION SICKNESS CAN APPEAR WITH THE DELAY AFTER THE AIRCRAFT IS ON GROUND

TUC (TIME OF USEFUL CONSCIOUSNESS)

TUC IS THE TIME UPTO WHICH PILOT CAN ACT PHYSICALLY AND MENTALLY WELL, AFTER TUC BOTH MENTAL AND PHYSICAL PERFORMANCE DEGRADE.

TUC VARIES INDIVIDUALLY AND DEPENDS ON CABIN PRESSURE ALTITUDE

ALTITUDE	TUC
18000 FT	30 MIN
22000 FT	5-10 MIN
25000 FT	3-5 MIN
30000 FT	1-2 MIN
35000 FT	30-60 SEC (30-90 SEC)
40000 FT	12 SEC (5-15 SEC)

— 40,000ft → 12 seconds.
— 35,000ft → 30-60 sec.
— 30,000ft → 1-2 min
— 25,000ft → 3-5 min
— 22,000ft → 5-10 min
— 18,000ft → 30 min

AFTER SCUBADIVING PILOT MAY DEVELOP SYMPTOMS OF DECOMPRESSION EVEN WHEN THE ALTITUDE IS LESS THAN 18000 FT

FLYING IMMEDIATELY AFTER SCUBA DIVING IS PROHIBITED. (SINCE IT INVOLVES RISK OF DECOMPRESSION SICKNESS)

FLYING AND DIVING

DO NOT FLY WITHIN 12 HOURS OF SWIMMING USING COMPRESSED AIR

AND AVOID FLYING FOR 24 HOURS IF A DEPTH OF 30 FEET HAS BEEN "

EXCEEDED.

What are the constituents of the atmosphere?

- a) Oxygen 22% Nitrogen 77% Other gases 1%
- b) Oxygen 22% Nitrogen 78% Other gases 2%
- ☒ c) Oxygen 21% Nitrogen 78% Other gases 1 %
- d) Oxygen 22% Nitrogen 77% Other gases 1 %

At what height is the partial pressure of oxygen in the lungs approximately half that at sea level?

- a) 10,000 feet
- b) 25,000 feet
- c) 30,000 feet
- ☒ d) 18,000 feet

What are the restrictions to flying after scuba diving?

- a) No flying within 48 hours if a depth of 40 feet has been exceeded, otherwise the limit is 12 hours.
- b) No flying within 48 hours if a depth of 30 feet has been exceeded, otherwise the limit is 24 hours.
- c) No flying within 12 hours if a depth of 30 feet has been reached, otherwise the limit is 24 hours.
- ☒ d) No flying within 24 hours if a depth of 30 feet has been exceeded, otherwise the limit is 12 hours.

Tidal volume is:

- a) The volume of air inhaled with each normal breath
- b) The volume of air exhaled with each normal breath
- c) The volume of air breathed when diving
- ☒ d) The volume of air inhaled and exhaled with each normal breath

EYE

☐ ***Light enters the eye through the Cornea***

☒ ***Light Adaptation. When experiencing sudden high levels of illumination the eye quickly adjusts in 10 seconds.***

☐ ***The most common factors affecting night vision are :***

a) Age

b) Mild hypoxia.

c) Cabin altitudes above 8,000 ft.

d) Smoking.

It takes time for our eyes to adapt to darkness. This adaptation does take time - about 7 minutes for the cones and 30 minutes for the rods.

R-30
C-7

VISUAL DEFECTS



Most visual defects are caused by the distorted shape of the eyeball.

1. ☐ Hypermetropia.

In long sightedness, Hypermetropia, a shorter than normal eyeball the image being formed behind the retina.

2. ☐ Myopia.

In short-sightedness, Myopia, the problem is that the eyeball is longer than normal and the image forms in front of the retina.

In the absence of anything to focus on (Empty field) the natural focus point of the eye is not at infinity, as was long assumed, but on average at a distance of between just under 1 metre and 1.5 metres

3. ☐ Cataracts.

Cataracts are nominally associated with the ageing process though some diseases can cause cataracts at any age. With time, the lens can become cloudy causing a marked loss of vision.

4. ☐ Glaucoma.

Glaucoma is a disease of the eye which causes a pressure rise of the liquid around the Eye

- ☐ Whereas the use of contact lenses by aircrew is permitted, under authorised medical supervision

Bi-focal contact lenses are prohibited .

COLOUR BLINDNESS

- **Total colour blindness is a bar to the issue of a flying licence. It is caused by a defect in the structure of the colour-sensitive cones in the retina**
- **It does not affect acuity and many people go through their lives with no knowledge that they suffer from this imperfection.**
- **It is the ability to discriminate at varying distances.**
- **DEFICIENCY OF VITAMIN A CAUSES NIGHT BLINDNESS, SINCE VITAMIN A IS ESSENTIAL FOR REGENERATION OF TISSUES**
- **ADAPTATION – THE ADJUSTMENT OF THE EYES TO HIGH OR LOW LEVELS ILLUMINATIONS**
- **BRIGHT LIGHT 10 SECONDS, DARKNESS 7 MIN FOR CONES, 30 MIN FOR RODS**
- **CONES ARE RESPONSIBLE FOR DAY VISION SINCE THEY ARE COLOUR SENSITIVE (CD TRICK)**
- **RODS ARE RESPONSIBLE FOR NIGHT VISION**
- **FOVEA – AREA IN WHICH CONES PRE DOMINATE**
- **FOVEA CENTRALIS – BEST FOR DAY VISION AND NO NIGHT VISION AT ALL**
- **RETINA IS A LIGHT SENSITIVE INNER LINING CONTAINING PHOTO RECEPTORS ESSENTIAL FOR VISION**
- **PERIPHERAL VISION IS IMPORTANT FOR DETECTING MOVING OBJECTS**
- **THE AMOUNT OF LIGHT ENTERING THE EYE IS CONTROLLED BY CORNEA**
- **THE AMOUNT OF LIGHT ENTERING THE RETINA IS CONTROLLED BY PUPIL**
- **PRESBYOPIA – FAR SIGHTEDNESS LINKED WITH AGE**

- **NIGHT VISION SCANNING : SCANNING TECHNIQUE IS TO LOOK FROM SIDE TO SIDE (15 – 20 DEGREES) OF AN OBJECT**
- **ACCOMODATION OF LIGHT IS DONE CRYSTALLINE LENS**

What is "empty field myopia"

- a) It is a term used when the eye, if it has nothing on which to focus will tend to focus at infinity.*
- b) It is a term used when the eye, if it has nothing on which to focus will tend to focus between 4 to 6 metres*
- c) It is a term used when the eye, if it has nothing on which to focus will tend to focus between 10 to 12 metres*
- d) It is a term used when the eye, if it has nothing on which to focus will tend to focus between just under 1 to 1.5 metres*

What causes long or short sightedness ?

- a) Presbyopia*
- b) Astigmatism*
- c) Distortion of the eyeball*
- d) Distortion of the cornea*

8. What four factors effect night vision?

- a) Age, alcohol, altitude and smoking*
- b) Age, altitude, Instrument lights and smoking*
- c) Instrument lights, alcohol, altitude and smoking*
- d) Age, alcohol, altitude and instrument lights*

Does lack of oxygen affect sight?

- a) Yes*

b) No

c) Sometimes

d) It depends on the health of the individual

Dark adaption takes about for the rods and for the cones

a) 30 minutes 7 minutes

b) 7 minutes 30 minutes

c) 15 minutes 20 minutes

d) 25 minutes 45 minutes

Glaucoma is caused by :

a) An increase in the pressure of the eye

b) A decrease in the pressure of the eye

c) A defect of the cornea

d) A defect of the retina

The oxygen required by the body is obtained from the air we breathe.

Whereas the brain only constitutes approximately 2% of body weight, it consumes 20% of the total required oxygen for the normal functioning of the body.

Tidal Volume is the volume of air inhaled and exhaled with each normal breath. It amounts to about 500 ml. in the normal male adult.

EAR

1. INNER EAR

CONSIST OF SEMI-CIRCULAR CANAL, OTOLITHS AND COCHLEA

2. MIDDLE EAR

CONSISTS OF TINY BONES, OSSICLES

INNER EAR IS RESPONSIBLE FOR PERCEPTION OF NOISE(COCHLEA)

BRAIN HAS TWO TYPES OF MEMORY.

1. Short-term memory → Also known as working memory it can remember about 7 items for about 30 Secs.

2. Working memory enables us to remember a clearance long enough to write it down.

3. The first phase of information process is sensory stimulation.

Long term memory an essential component of pilot's knowledge and expertise. It is desirable to pre-activate knowledge stored in long term memory to have it available when required.

Shes occur whenever a pilot revise his plan of action and does not immediately have a Solⁿ.

A moderate level of shes may improve performance

shes promotes ~~or~~ physical strength rather than mental performance.

Symptoms of overstrained pilots:—

1. Mental block → Confusion → Frustration → Deterioration in motor Coordination.

70-80% of Aviation Accidents are caused by Human factors rate of Accident in Commercial Aviation are one Accident/million flight movements.

VESTIBULAR APPARATUS (OTOLITHS + SEMICIRCULAR CANALS) HELPS MAINTAINS SPATIAL ORIENTATION

OTOLITH SENSES LINEAR ACCELERATION

SEMICIRCULAR CANAL SENSES ANGULAR ACCELERATION

INTENSITY OF SOUND IS MEASURED IN DECIBELS

HUMAN HEARING RANGE - 16 HZ – 20000HZ

16 HZ – 20 KHZ

PROLONGED EXPOSURE TO NOISE IN EXCESS OF 90 DECIBELS CAN END UP IN NOISE INDUCED HEARING LOSS

BAROTRAUMA – IS LIKELY TO CAUSE PAIN CAUSED BY DIFFERENCE IN PRESSURE EXISTING BETWEEN THE MIDDLE EAR AND THE AMBIENT AIR (EITHER SIDE OF EAR DRUM)

SOLUTION – REDUCE RATE OF DESCENT OR CLIMB OR LEVELLING OUT , GIVE MORE TIME FOR PRESSURE BALANCING

Illusions

Vertigo – Another illusion associated with the vestibular apparatus is Vertigo, a loss of spatial awareness, in which the individual experiences a rotating, tumbling or turning *Because of Rapid Head movement.* sensation.

Coriolis Effect

If a steady turn is being maintained then a sudden movement (greater than about 30 a second) of the head will be detected as a change in the turn rate.

Somatograval Illusion - The Somatograval Illusion results from the inability of the semi-circular canals to register accurately a prolonged rotation

Thus a Somatograval Illusion is the sensation of turning in the

Seas of pants :- it is the illusion caused by movement of body parts and muscle or joints.

Seas of pants Sense :- The pants Sense can give false inputs to the body orientation when visual reference is lost.

Respiration System :-

TIDAL VOLUME (500ml) :- it is the Vol. of air inhaled and Exhaled with each normal breathe.

Normal rate of Breathing of an Adult is at rest is about 16 cycles/min or 12-20 Cycles/min.

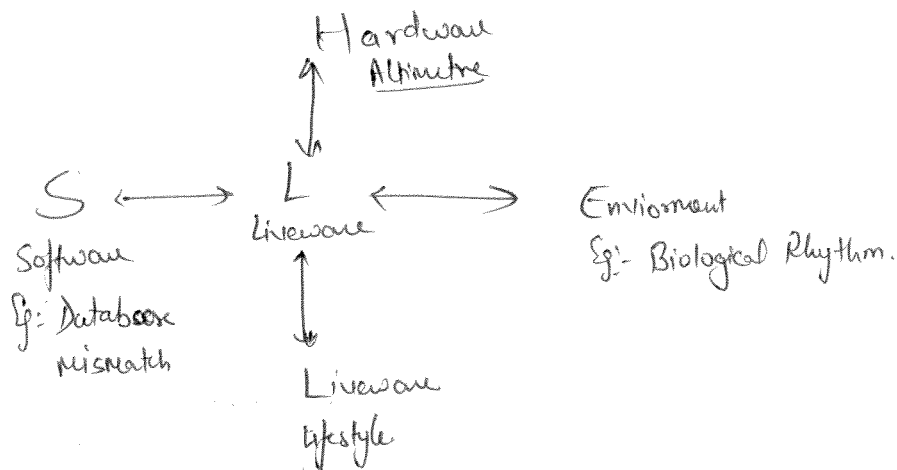
The transfer of O_2 from the Alveoli to the Blood can be described by Law of Diffusion.

Pulmonary Artery :- Oxygen poor CO_2 rich Blood.

The part of Blood without cells is plasma.

Cardiac output = 5 L/min.

Average pulse of Healthy Adult = 60-80 Beats/min



opposite direction that occurs whenever the body undergoes angular deceleration from a condition of sustained angular velocity.

Somatogravic illusion The illusion of pitching up or down as a result of the movement of the Otoliths due to linear acceleration

If IMC : BELIEVE YOUR INSTRUMENTS

If VMC : LOOK OUT AT THE HORIZON

AUTOKINETIC ILLUSION – (AUTOKINESIS) ILLUSION IN WHICH A STATIONARY LIGHT IF STARED FOR SEVERAL SECONDS IN THE DARK WITHOUT ANY VISUAL REFERENCE APPEARS TO MOVE.

NARROW RUNWAY – PILOT MAY FEEL HE IS AT A GREATER HEIGHT THEN ACTUAL AND TENDENCY TO LAND SHORT. (UNDERSHOOT) LAND SHORT (LATE round out)

WIDE RUNWAY – PILOT MAY FEEL HE IS LOWER THAN ACTUAL AND HAS A TENDENCY TO OVERSHOOT THE RUNWAY

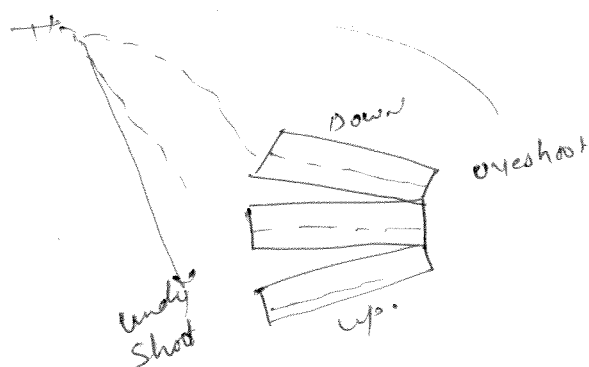
RUNWAY SLOPE

DOWNSLOPE - HE MAY FEEL HE IS LOWER THAN ACTUAL, THIS ILLUSION MAY CAUSE HIM TO LAND AHEAD (HIGH ROUND OUT) Undershoot

UPSLOPE - MAY FEEL HE IS HIGHER THEN ACTUAL, THIS ILLUSION MAY CAUSE HIM TO LAND SHORT overshoot

SEATS OF PANTS - Caused by movement of body parts or muscle or joints.

SEATS OF PANTS SENSE - pants sense give false inputs to body orientation when visual reference is lost



THE AIRCRAFT ACT 1934 AND AIRCRAFT RULES 1937

THIS ACT WAS PASSED BY THE INDIAN LEGISLATORS AND RECEIVED THE ASSENT OF THE GOVERNOR GENERAL ON 19TH AUGUST 1934 . IT IS THEREFORE KNOWN AS THE AIRCRAFT ACT 1934.

IT EXTENDS TO THE WHOLE OF INDIA AND APPLIES –

- A. TO CITIZENS OF INDIA WHERE EVER THEY MAY BE
- B. TO THE PERSON ON, AIRCRAFT REGISTERED IN INDIA WHEREEVER THEY MAY BE .

THE ACT OF 1934 AND RULES FRAMED IN 1937 WERE BROUGHT OUT IN A BOOK WHICH WAS KNOWN AS THE INDIAN AIRCRAFT MANUAL.

THIS MANUAL IS PUBLISHED EVERY FOURTH YEAR BY THE OFFICE OF DGCA.

THE NAME OF THIS BOOK HAVE BEEN CHANGED TO AIRCRAFT MANUAL.

THE ACT 1934

- ✓ 1. PROHIBITION OF SLAUTERING OF ANIMALS AND DEPOSITING RUBBISH WITHIN A RADIUS OF 10 KM FROM THE ARP.
- ✓ 2. POWER TO DETAIN ANY A/C BY CENTRAL GOVERNMENT
- ✓ 3. POWER TO PROHIBIT THE CONSTRUCTION OF BUILDINGS AND , PLANTING OF TREES WITHIN 20 KM OF THE ARP.
- ✓ 4. PENALTY FOR FLYING SO AS TO CAUSE DANGER –2 YEARS IMPRISONMENT OR 10 LAKH FINE OR BOTH
- ✓ 5. REMOVAL OF DAMAGE A/C

A. ON A/C WRECKED ON WATER OR LAND MAY BE REMOVED UNDER THE SUPERVISION OF OFFICER OF POLICE OR MAGISTRATE OR CAD(CIVIL AVIATION DEPARTMENT)

LANDING, PARKING AND HOUSING CHARGES ARE BASED ON THE TOTAL ALL UP WEGHT (AUW) OF AN AIRCRAFT. CONTAIN IN THR C OF A.

Smoking permitted if written in Certificate of Airworthiness.

THE AIRCRAFT RULES 1937

- ✓ 1. CLASS RATING – SINGLE ENGINE LAND, SINGLE ENGINE SEA, MULTI ENGINE LAND AND MULTI ENGINE SEA.
- ✓ 2. PETROLEUM IN BULK – PETROL CONTAIN IN AMOUNT OF EXCEEDING 900 LITRES IN CAPACITY.
- ✓ 3. TYPE RATING – RATING ON EACH AIRCRAFT WHENEVER CONSIDERED NECEAASRY BY THE AUTHORITY.

4. STATE AIRCRAFT – INCLUDE MILITARY AIRCRAFT AND A/C EMPLOYED FOR GOVERNMENT SERVICES EX- POLICE, CUSTOMS, POST ETC
5. MILITARY AIRCRAFT – INCLUDE AIRFORCE MILITARY AND NAVAL AIRCRAFTS
6. INSTRUMENT TIME – INSTRUMENT GROUND TIME + INSTRUMENT FLIGHT TIME
7. EAT – EXPECTED APPROACH TIME – THE LASTEST TIME WHEN THE AIRCRAFT IS EXPECTED TO LEAVE THE HOLD ~~AFTER~~ ^{Simultaneous} COMPLETING THE INSTRUMENT APPROACH FOR FINAL DESEND FOR LANDING.
8. ETA- ESTIMATED TIME OF ARRIVAL – THE TIME A/C IS ESTIMATED TO REPORT OVERHEAD STATION OR AT IAF.

AIRCRAFT PUBLIC HEALTH RULES 1954 –

THE COMMANDER OF THE A/C MUST INFORM THE HEALTH OFFICER IN INDIA ATLEAST 2 HOURS IN ADVANCE ON R/T OF ANY PERSON SUFFERING FROM QUARANTINABLE DISEASES.

INCUBATION PERIOD –

CHOLERA	CO- 5 DAYS
PLAGUE	P – 6 DAYS
YELLOW FEVER	Y – 6 DAYS
RELAPSING FEVER	RE – 8 DAYS
SMALL POX	S – 14 DAYS
TYPHUS	T – 14 DAYS

Colera → Plague → Relapsing fever
 CO PY RE S T → Typhus.
 5 6 6 8 14 14.
 yellow fever Small pox
 CO PY RE ST

^{Ashes.}
 CARRIAGE OF DEAD BODIED AND CREMATED REMAINS
 DEAD BODY OR HUMAN REMAINS THOSE WHO DIED BECAUSE OF THE
 THE FOLLOWING CANNOT BE BROUGHT TO INDIA.

PAYG → ^{plague} Plague, Anthrax, Yellow fever, Cholera.

PROVIDED CREMATED ASHES OF DEADBODIES OR HUMAN REMAINS CAN BE BROUGHT TO INDIA.

CREMATED ASHES SHALL BE PLACED IN AN URN OR CASKET HAVING AN OUTER PACKING OF SUITABLE MATERIAL.

IN CASE OF UNCREMATED REMAINS – IT SHOULD BE HERMETICALLY SEALED IN A ZINC CONTAINER.

AIR DEFENCE CLEARANCE

ADC IS TO ISSUED EXCEPT WHEN FLYING WITH IN A RADIUS OF 5 NM FROM THE ARP. AND VERTICAL LIMIT OF 1000 FT (NO ADC REQUIRED)

ADC VALIDITY

ADC IS VALID FOR THE ENTIRE ROUTE IRRESPECTIVE OF NUMBER OF HOLTS IN BETWEEN

MORE THAN 30 MIN DELAY IN DEPARTURE NEW ADC IS TO BE REQUESTED

ALL AIRCRAFT NEED ADC 10 MIN BEFORE ENTERING ADIZ(AIR DEFENCE IDENTIFICATION ZONE)

CARRIAGE OF CABIN CREW

No of Seats

10-49	1
50-99	2
100-149	3
150-199	4
200-249	5
250-299	6

CARRIAGE OF ARMS AND EXPLOSIVE (DANGEROUS GOODS)

ONLY WITH THE WRITTEN PERMISSION IN WRITING WITH THE CENTRAL GOVERNMENT

CARRIAGE OF MAILS

ONLY WITH THE PERMISSION OF DGPT DIRECTOR GENERAL OF POST AND TELEGRAPH

CARRIAGE OF PRISONERS

WITH PERMISSION OF DGCA, DDG (DEPUTY DIRECTOR GENERAL) , DRI (DIRECTOR OF REGULATION AND INFORMATION), CAD (CIVIL AVIATION DEPARTMENT)

CARRIAGE OF PERSON SUFFERING FROM MENTAL DISORDERS OR FROM EPILEPSY – ONLY WHEN ACCOMPAINED BY THE MEDICAL PRACTITIONER

PHOTOGRAPHY AT AN AERODROME CAN BE DONE WITH THE PERMISSION OF DGCA.

OPERATIONAL MANUAL IS PREPARED BY THE OPERATOR(JET AIRWAYS) AND APPROVED BY DGCA.

REFUELLING

B BUILDING – 15 M
P PERSON – 15 M
S SMOKING – 30 M (NAKED FLAME)
J JET BLAST – 43 M (200 FT)

B P S J
15 15 30 43

MAXIMUM SPEED FOR CIVIL AIRCRAFT .92 MACH

FDTL [FLIGHT DUTY TIME LIMITATIONS.]

DAY → 8 HRS.

WEEK → 35 HRS.

MONTH → 125 HRS as PIC / 100 PIC & 30 as Co-PIC

YEAR → 1000 HRS.

	FLY TIME	FLIGHT DUTY TIME	No. of landings
2 crew	8 HRS. / 9 HRS.	11 HRS.	6 / 3 or 2 in flight.
2 + 1 crew = 3 crew	9 HRS.	12 HRS.	6
2 crew	9 HRS. / 10 HRS.	13.	3 / 1
3 crew usually two on the	10 HRS.	12.	3.

FDTL Extension.

FLY TIME. FLY DUTY TIME
 $1\frac{1}{2}$ hr. 3 HRS.
 3 HRS. 6 HRS. in
 in 30 days consecutive 30 days.

① → flight time can be extended by maximum of 1 and half hour. flight duty time can be extended Max. of 3 HRS. Subject to a cumulative of 3 HRS of flying time and 6 HRS. of flight duty time in 30 consecutive days.

② → Every aircraft must carry checklist, emergency checklist, Route guides, ATC log book.

Max. Extension of flight time is

1.5 HRS per day. In a cumulative limit of 3 HRS in 30 consecutive days.

CVR (COCKPIT VOICE RECORDER) RECORDS 30 MIN/2 HOURS OF CREW CONVERSATION HELPS IN INVESTIGATION

FDR (FLIGHT DATA RECORDER) – IT RECORDS FLIGHT DATA FOR 25 HOURS LOCATED BACK SIDE OF THE AIRCRAFT HELPS TO KNOW A/C PERFORMANCE AND INVESTIGATION

COLOUR OF FDR IS ORANGE

DOCUMENTS TO BE CARRIED ON BOARD

1. **CERTIFICATE OF REGISTRATION**
2. **CERTIFICATE OF AIRWORTHINESS**
3. **CREW FLIGHT LICENCE**
4. **RADIO LICENCE**
5. **PASSENGER LIST**
6. **CARGO MANIFEST**
7. **LOGBOOK AIRCRAFT**

CERTIFICATE OF REGISTRATION IS VALID TILL THE TIME AIRCRAFT IS DESTROYED IN AN ACCIDENT.

SMOKING IS PERMITTED OR PROHIBITED IS WRITTEN IN THE C OF A

CREW LOG BOOKS SHOULD BE PRESERVED FOR 5 YEARS FROM THE DATE OF LAST ENTRY

DROPPING OF PAPER LEAFS OR FLOWERS FROM AIR IS DONE WITH THE PERMISSION OF LOCAL DISTRICT MAGISTRATE.

SUCCESSION OF COMMAND IS GIVEN BY OPERATOR

THE COLOUR IDENTIFICATION OF DROPABLE CONTAINERS

1. **RED FOR MEDICAL SUPPLIES AND FIRST AID EQUIPMENTS**
2. **BLUE FOR FOOD AND WATER**
3. **YELLOW IS FOR BLANKET AND PROTECTIVE CLOTHING**
4. **BLACK FOR MISCELLANEOUS EQUIPMENT**

GROUND AIR VISUAL SIGNALS

V- REQUIRE ASSISTANCE

X- REQUIRE MEDICAL ASSISTANCE

SOS – SAVE OUR SOULS

*for flight duty line
3hrs/day to a limit of
6hrs in 30 days.*

SAHIL KHURANA
SEC – 7, DWARKA, NEW DELHI
09871866290

Fire.

- Class A-fire - organic material such as paper, cotton etc.
Use $\rightarrow$ H_2O , CO_2 , dry chemical, halon (bif),
bromo chloro difluoro methane
- Class B-fire $\rightarrow$ liquid fire $\rightarrow$ fuel fire ~~the~~
Use $\rightarrow$ CO_2 , Halon, Dry chemical Except Water
- Class C-fire $\rightarrow$ liquified gaseous fire $\rightarrow$ Use CO_2 , Dry powder, Halon
- Class D-fire $\rightarrow$ Metal fire, landing gear fire,
Use $\rightarrow$ Halon, Dry chemical, Sand except CO_2 and Water
- Flight Deck $\rightarrow$ Halon and CO_2 Can be used.
- Brake fire $\rightarrow$ Dry powder and Sand can be used.
- CO_2 $\rightarrow$ is good for fire, plastic, fuel, electric fire.
But should not be used for metal fire.
- water Glycol should ~~be~~ not be used for electric fire.
 $\rightarrow$ electric fire $\rightarrow$ CO_2 and BCF, water Glycol should never be used.
- $\Rightarrow$ Engine fire $\rightarrow$ Halon, BCF, methyl Bromide
- Cabin here $\rightarrow$ water Glycol, BCF
- Brake fire $\rightarrow$ O_2 is never used on brake fire

OXYGEN REQUIREMENTS

620 hpa.

1. **REQUIRED FOR ALL WHEN FLYING ABOVE 13000 FT ALL THE TIME**
2. **FROM 10000-13000 AFTER 30 MIN O2 IS REQUIRED FOR ALL THE FLIGHT CREW MEMBERS AND 10% OF PASSENGERS**

FUEL REQUIREMENTS

PISTON PROP AIRCRAFT	TURBOJET AIRCRAFT (TURBINE ENGINE)
WHEN ALTN AERODROME IS REQUIRED (IFR FLIGHT) FUEL TO DSTN + FUEL TO ALTN + 45 MIN OF HOLDING FUEL	WHEN ALTN AERODROME IS REQUIRED (IFR FLIGHT) FUEL TO DSTN + FUEL TO ALTN + 30 MIN OF HOLDING FUEL OVER ALTERNATE AERODROME @ 1500 FT AGL
WHEN ALTN IS NOT REQUIRED (VFR FLIGHT) FUEL TO DSTN + 45 MIN OF HOLDING FUEL	WHEN ALTN IS NOT REQUIRED (VFR FLIGHT) FUEL TO DSTN + 30 MIN OF HOLDING FUEL @ 1500 FT AGL
WHEN ALTN IS NOT AVIALABLE (ISOLATED AERODROME) - FUEL TO DSTN + 45 MIN OF FUEL AND 15 % OF TRIP FUEL - OR 2 HOURS OF FUEL AT NORMAL CONSUMPTION (WHICH EVER IS LESS)	WHEN ALTN IS NOT AVIALABLE (ISOLATED AERODROME) FUEL TO DSTN + 2 HOURS OF FUEL AT NORMAL CONSUMPTION

FIRST AID KITS.

Seats.

1-100 → 1

100-200 → 2

200-300 → 3

no. of fire extinguishers.

Seats.

7-30 → 1

31-60 → 2

60-200 → 3

200-300 → 4

300-400 → 5

H.W 14/7/15
Pg-53 1-25 1-40.
Pg 334, 335, 336.

Test on Saturday

- PANS
- Radar separation
- Aerodromes and lights
- Personal Licensing.

H.W Human Performance
Repeat Pg-333 para.-336.
Repeat Pg-53 Aerodromes.

H.W. on 23/11.
ICAO Convention 20

H.W 27/07/15.

H.P Complete All question
BB &c

H.W 28/07/15.

Sample paper - 1 BB.
Pg 238 BB.

five and made.
sample paper - 3.

Coming Saturday test

@ Airspan - Rules of Air
and AIS

Why of fuel does not directly add?
4/10 pilot?

Pg-145 1-45
S.P-2

Pg No → 199

Test on Saturday.

Human Performance

National Law

Rules of Air

AIS

1	2	3	4
5	6	7	8
9	10		

H.W. Pg-145, 146, 147. H.P.

- 1) X.A
- 2) D
- 3) DX A
- 4) D

Pg-149 Rules of Air
Qs 1-30

H.W BB.

Pg-70 Qs 1-26.

23/9/15

Pg-155 41-70
Pg-104 1-60.

~~Test~~ - H.W 30/9/15.

Pg-291, BB, Qs 1-30

Pg 16 → 53 Part. 2

chapter 17 1-30 Aerodromes.

Revise Law
weakest chapter

Pg-321 Sample 10 → 1.